What a Disaster Means: Attention, Interpretation, and the Demand for Green Assets

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Abstract

How events are interpreted locally substantially shapes portfolio choices, fund flows, and asset prices. In the setting of extreme weather events, I use fine-tuned large language models on 1.5 million articles from U.S. news outlets to construct a new measure of local event interpretation that distinguishes attention to a disaster from coverage explicitly linking it to climate change. I combine this measure with geographically identified portfolios of household-serving advisers and mutual fund managers. Among disaster-affected counties, stronger climate attribution leads to substantial reallocation toward green ETFs in household portfolios. The relationship extends to ESG mutual-fund inflows and to nearby fund managers, who tilt their active trading toward greener firms as climate-linked coverage rises. Predetermined county and investor characteristics and the direct economic effects of the disaster are not driving the responses. Using changes in the ownership structure of the county's newspapers, I provide causal evidence that locally supplied climate interpretation narratives move households' and fund managers' portfolios independently of readers' prior beliefs and preferences. In a calibrated model where institutional investors respond to one of the many disasters occurring each year, an interquartile increase in local climate interpretation implies a relative appreciation of green stocks equivalent to up to 15.2% of the observed annual green return spread.

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1 Introduction

Scheduled firm announcements account for a small share of the information that moves stock prices (Ball and Shivakumar, 2008). Much of the rest comes from events that are not objectively and unambiguously connected to any single firm, such as tariffs, a pandemic, an election, a new technology or a hurricane. How such an event affects a stock depends on what the investor takes it to mean for that stock, and different investors read the same event differently (Kandel and Pearson, 1995; Cookson, Engelberg, and Mullins, 2020; Meeuwis et al., 2022; Kempf et al., 2023).[1] It is hard to study whether this reading shapes portfolio choices, because across events the content, the assets affected and the speed of the reaction also differ.[2]

In this paper, I study how the interpretation of an event affects the direction in which investors adjust their portfolios. I examine whether households and fund managers who experience a disaster move toward sustainable assets when their local news reads it as evidence of climate change, comparing investors exposed to the same disaster and to the same amount of coverage of it. I also examine whether this response reflects their beliefs about climate risk or their politics and incentives.

Climate change has increased the frequency and intensity of several weather extremes, including heatwaves and heavy precipitation in many regions (IPCC, 2021). Extreme weather events let me compare how the same event is read in different places. A flood or hurricane hits many counties at the same time, each with its own investors and its own local information environment, and whether it is a manifestation of climate change is contested. After Hurricane Ian, the share of respondents associating the storm with climate change differed by 21 to 41 percentage points between liberals and conservatives, and liberal outlets were more likely to suggest the link and conservative outlets to deny it (Djourelova et al., 2024). A single episode says little on its own about the climate trend (Choi, Gao, and Jiang, 2020), so what investors do after it depends on how they interpret it. Managers near a disaster reduce their holdings of the firms exposed to it (Alok, Kumar, and Wermers, 2020), treating the event as a local hazard. If an investor takes the same event as corroborating evidence of climate change, it might also concern sustainable investments, a category that held $3.9 trillion at the end of 2025 (Morningstar, 2026) and includes firms and places far from the damage. Outside labeled funds, ESG-related tilts total about 6% of the investment industry's assets from 2012 to 2023 and rise from 17% to 27% of institutions' total portfolio tilts (Pástor, Stambaugh, and Taylor, 2025).

I show that the same physical event can receive different interpretations across affected places and use this variation to study portfolio reallocation by geographically identified institutional investors. I train an ensemble of large language models and use them to classify nearly 1.5 million articles from 527 U.S. news outlets. I combine these classifications with ZIP- and county-level newspaper circulation data, reaching 2,134 counties, to construct a novel measure of national and local interpretation, validated to separately identify simple attention to a disaster from coverage explicitly interpreting the event as evidence or a manifestation of climate change. I study two settings, local investment advisers serving household clients and actively managed mutual funds. Both let me observe the portfolio adjustment of an individual investor in a known county and relate it to the local coverage of each disaster and, for funds, to the political donations of the investment team.

Comparing counties affected by hurricanes and floods, I find that for portfolios managed by local advisers

[1] For example, the 2024 election could be read as bad news for electric vehicles or as good news for Tesla relative to its competitors, and DeepSeek's cheaper AI model as lower or higher demand for Nvidia's chips.

[2] Which firms an event reaches and how fast investors price it are not given by the event: after the 2016 election, investors incorporated the growth and tax consequences for firms quickly and took longer to work out the trade consequences (Wagner, Zeckhauser, and Ziegler, 2018).

serving household clients, a ten-percentage-point higher share of local news coverage explicitly linking the disaster to climate change is associated with a 19-basis-point larger increase in the green-ETF share over two quarters, about $5 million reallocated to green ETFs at the average county ETF portfolio size, or 19.6% of average pre-event green-ETF holdings. Higher disaster coverage and attention alone lead to greater overall turnover, but it is the interpretation of the disaster as related to climate change that leads to reallocation specifically toward green assets. The relationship extends from advisers' green ETFs to flows into ESG mutual funds: among funds headquartered in affected states, net inflows into ESG funds relative to non-ESG funds are larger after disasters with high climate interpretation, by 8.5 percentage points of pre-disaster assets over two quarters and 18.6 over three, with no such difference in neighboring unaffected states.

The challenge in identifying this response is that a disaster can move green holdings without any attribution to climate change. Disasters can strike places where climate concern and green investment are already rising. The damage can create liquidity needs, and the attention a disaster receives can move portfolios whether or not its coverage relates it to climate change. Advisers' own views and incentives can move the portfolios they manage for households. I compare changes in portfolios before and after a disaster between affected counties and clean comparison counties around the same episode, and within the same disaster I compare investors at the same distance from it whose local coverage reads it more or less as climate change, conditional on disaster attention. The response is measured at pre-event prices, net of valuation changes. Pre-disaster changes in the green share do not differ between affected and comparison counties, and the estimate does not change when I control for local trends and non-disaster climate attention. It is not explained by the county's prior climate concern, political orientation, income or education. Green ETF assets rise together with non-green ETF assets while total holdings do not change, so liquidation does not produce it. The local response does not change when I add the national coverage and the national climate interpretation of the same disaster, and it is similar after years of strong and weak green-ETF returns. The response varies with the political composition of the county and not with the adviser's own political orientation.

Across outlet–episode pairs, stable differences between outlets, such as the political views and green attitudes of their audiences and owners, account for 5.5% of the variance of local climate interpretation, the characteristics of the event for 40.7%, and how a given outlet reads a given disaster for 54.1%. The response to interpretation is the same in counties with more and fewer residents worried about global warming, and does not vary with the county's perceived scientific consensus, Democratic vote share or college share.

The interpretation measure already captures the reading of the event that forms locally, whether the county's outlets supply it or cater to their readers' predisposition. The estimates measure the response to local interpretation from either source. I use event-study pre-trends, earthquake placebos, controls for the county's politics, climate beliefs and demographics, and fixed effects absorbing the county's trends and usual response to disasters to rule out that the response comes from county trends or a general propensity of climate-concerned counties to buy green assets after any salient local shock. Moreover, to further investigate the role of locally supplied models and narratives (Schwartzstein and Sunderam, 2021), I use supply-side variation in interpretation. Many local newspapers belong to chains, which supply centrally produced copy, and under a larger owner the climate-interpreted share of local disaster coverage is lower. The instrument is the size of the chain that owned the county's outlets before the event. It varies at sell-offs, purchases and mergers of those outlets, as well as upstream mergers and acquisitions of their owners.[3] Changes in chain size do not predict the county's vote share, its climate beliefs, its portfolio change before the event,

[3]For example, the 2019 merger of Gannett and GateHouse.

or the losses and coverage of the disaster. Instead, these transactions follow declining advertising revenue, the liquidity needs of legacy owners and cost cutting in roll-ups (Ewens et al., 2022; Gao et al., 2020). I compare the county's response to disasters of the same type before and after the chain size of its newspapers changes. Per ten percentage points of the interpretation share, the instrumented response is 22 to 25 basis points for counties and 19 to 23 basis points for advisers, in line with the local-projection estimates.

To further close the gap between the local interpretation an investor is exposed to and the beliefs of the investor who makes the decision, I turn to actively managed mutual funds, where I observe the investment team that makes the allocation decision directly. From the political donations of the managers and investment teams responsible for the portfolio, I measure their prior views. The office where the portfolio is run gives the local interpretation each team is exposed to. I also separate active stock reallocation from the proportional adjustment of holdings to investor flows, which isolates the managers' own decisions from those of fund investors. I focus on funds without an ESG mandate, to rule out rebalancing that follows contractual requirements, incentives and considerations unrelated to local disasters. As a result, the active rebalancing induced by interpretation can be read as a choice by the managers, driven, for example, by belief revision on climate risk. I find that a ten-percentage-point higher share of climate-linked coverage reaching managers within 50 miles of a disaster raises their active tilt toward greener firms by roughly 70 basis points in the quarter of the disaster, while the volume of disaster coverage alone does not move the tilt. The response is also robust to supply-induced variation in interpretation, using the same instrumental variable as for advisers. The response depends on where the managers work and not on where the fund is headquartered. Managers near the event reallocate even when the headquarters is far from it, and funds headquartered near the event do not respond when their managers work elsewhere. As for advisers, the response declines with distance from the event. Detailed holdings also let me compare how funds change their weight in the same stock. Fixed effects for each stock around each disaster, within each fund style, absorb the stock's return and fundamentals, the location of the firm and its exposure to the disaster, and the shock to its industry. They also remove differences in the kinds of stocks held by managers with different climate beliefs or incentives. The comparison is then between managers who hold the same stock and whose local coverage reads the event differently.

I then provide suggestive evidence on the mechanism, using survey responses and insurance purchases. In survey data, expectations of imminent or ongoing harm from global warming are positively associated with local advisers' green ETF holdings and rise after floods among moderates and conservatives, with weaker evidence for beliefs about human causation and for policy attitudes. Non-flood extreme weather events predict stronger flood-insurance take-up in affected counties only where local newspapers link them to climate change, a cross-hazard response that salience of the experienced hazard alone does not produce. Together, these patterns are consistent with disaster experience raising perceived climate harm where local news interprets the disaster as climate change, and with perceived harm leading to green investment.

At the national level, noteworthy national disasters raise flows into green funds and the relative returns of sustainable assets (Marshall et al., 2021; Fiordelisi et al., 2023; Misev and Balles, 2024). The local responses above can move prices when many investors read a disaster the same way at the same time. The climate interpretation of national disaster coverage, beyond the attention a disaster receives, is associated with flows into ESG funds. At the average national interpretation share of 25.6%, a one-standard-deviation national disaster-attention surprise is associated with an estimated cumulative increase in flows into ESG-labeled funds relative to other funds equivalent to approximately $288 million over the attention month and the following two months. The estimated three-month response is approximately $580 million larger at the

75th than at the 25th percentile of the national interpretation share. Shifts in institutional ESG demand account for about 86% of the average annual return on a long-short ESG portfolio (van der Beck, 2026). To gauge the economic significance of interpretation, I consider a counterfactual in which disasters throughout a year collectively activate the institutional investor base, with each investor responding once as the local households represented in my RIA sample do. Holding attention fixed, an interquartile increase in climate framing implies roughly 34 basis points of one-time ESG-stock appreciation relative to other stocks, using multipliers calibrated from van der Beck's (2026) model and assuming comparable portfolio demand and market accommodation. This is equivalent in magnitude to 18.2% of the average annual return contribution attributed to institutional ESG demand and 15.2% of the observed annual return spread. Evidence of similar effects of interpretation on active stock reallocation by mutual-fund managers motivates extending the exercise from flows into labeled ESG funds to institutional equity demand.

Disagreement, political views, and financial responses to common events. This paper contributes to research on disagreement, political views and financial responses to common events, which documents partisan differences in investment beliefs and capital allocation (Meeuwis et al., 2022; Kempf et al., 2023; Cookson, Engelberg, and Mullins, 2020). I study the effect of local narratives and interpretations beyond political views, and how the two interact. My constructed measure captures geographic differences and changes over time in the explanations linking experience to climate change, as well as shared interpretations of specific events that more stable, cross-sectional differences in political views alone cannot capture.

Disasters, climate attention and financial markets. The paper also relates to research linking climate attention to trading and asset returns (Choi, Gao, and Jiang, 2020; Pástor, Stambaugh, and Taylor, 2022; Ardia et al., 2023).[4] I contribute by identifying local portfolio responses to disaster experience and examining how these responses depend on whether local narratives attribute the event to climate change. I classify whether each local news article presents a particular disaster as evidence of climate change, assign that coverage to investors' counties through newspaper circulation, and measure it separately from the attention the disaster receives. I then trace how investors in different places adjust their portfolios to the same disaster and its local coverage. Disaster coverage is associated with greater gross turnover, while climate attribution is associated with reallocation toward green assets.

I further contribute to research on institutional investors' responses to climate-related shocks. Prior work documents reduced exposure to disaster-zone stocks, greater support for environmental proposals, and portfolio responses to climate-related belief shocks (Alok, Kumar, and Wermers, 2020; Fich and Xu, 2025; Alekseev et al., forthcoming).[5] I examine how the direction of portfolio adjustment varies with political orientation and the extent to which local narratives attribute a disaster to climate change. Holding disaster attention fixed, I show how stronger attribution is associated with active reallocation toward greener assets.

Surveys and portfolio holdings. Finally, the paper relates to research linking investors' beliefs and pref-

[4]Choi, Gao, and Jiang (2020) associate unusually warm local temperatures with increased attention to climate change, retail selling of carbon-intensive stocks, and their relative underperformance. Pástor, Stambaugh, and Taylor (2022) link green assets' realized outperformance to unexpectedly strengthening climate concerns. Ardia et al. (2023) construct a daily newspaper-based climate-concern index and show that green stocks outperform brown stocks on days when climate concerns rise unexpectedly. Their daily stock-return regressions also associate green outperformance with shocks to concerns about specific physical risks, including hurricanes and floods.

[5]Alok, Kumar, and Wermers (2020) find that funds headquartered within 100 miles of a disaster-zone firm reduce their portfolio weight in that firm more than distant funds, with evidence consistent with salience bias. Alekseev et al. (forthcoming) use local extreme heat and changes in shareholder disclosures to identify climate-belief shocks and connect them to industry reallocations, like buying automobile and semiconductor stocks and selling real estate. Fich and Xu (2025) find increased support for environmental proposals after hurricanes and increased holdings in firms where their votes could help environmental proposals pass, providing evidence consistent with both managers' own preferences and career concerns about investor flows.

erences to observed portfolio holdings (Riedl and Smeets, 2017; Giglio et al., 2025; Anderson and Robinson, 2026). While I do not link individual survey responses directly to their portfolio holdings, I contribute to this evidence on cross-sectional heterogeneity in green ownership by examining portfolio adjustments around particular disaster experiences instead, using pre-event allocations to distinguish investment responses from pre-existing differences in holdings.[6] Investors rarely adjust their portfolios: changes in beliefs explain little about whether they trade, even though they help explain the direction and magnitude of trading conditional on adjustment (Giglio et al., 2021). The reallocations I observe suggest that disaster experience and its local interpretation can overcome this inertia.

Experience, media, and climate attitudes. The paper is related to a broader literature on how experience, political views, and media shape climate attitudes (Deryugina, 2013; Ash et al., 2023; Djourelova et al., 2024). My contribution is to document how the local interpretation of disaster experience is reflected in financial decisions, specifically through active portfolio reallocations toward green assets. Complementary survey evidence also examines how responses to disaster experience vary with political views and local media coverage, distinguishing perceived climate risks from beliefs about human causation and policy preferences. These patterns suggest that risk perceptions can respond to experience among conservatives despite persistent political differences in climate attitudes.[7] Read alongside the portfolio responses, they point to the role of perceived risk in prompting green investment adjustments.

Section 2 describes the data, the institutional setting, and the measurement of climate interpretation. Section 2.2 characterizes the local information and interpretation environment and the determinants of interpretation.

I investigate two separate channels through which climate interpretation can operate. In Section 3, I study household demand for green ETFs through investment advisers. Given the low elasticity to prices of this investor group, and since green ETFs are mostly passive indexes, this channel is meant to isolate the effect of interpretation on non-fundamental demand. I extend the analysis to ESG mutual-fund flows. Section 3.3 discusses the specific confounders each set of controls rules out. In Section 3.4, I address and rule out the key alternative explanations and concerns pertaining to (1) whether the response is induced by interpretation or by attention, predisposition, climate change salience or another confounder, (2) whether the response is tied to a specific event, and (3) whether changes in green ETF demand reflect demand shocks for sustainability. In Section 4, I study active rebalancing by mutual-fund managers.

In Section 5, I test whether the local interpretation moves portfolios when it is driven by changes in the narrative that the outlets supply, independently of the demand or preferences of their readers. In Section 6, I elaborate on concern (3), showing the main determinants of the green demand shock. Climate interpretation of disaster events can affect both flows and prices, depending on the identity and size of the investors who are activated and of who accommodates. In Section 7, I relate the climate interpretation of national disaster coverage to flows into ESG funds. I then quantify the appreciation of green stocks implied by an interquartile increase in climate framing if a year's disasters collectively activate the institutional investor base. Section 8 concludes.

[6] Closest to my paper, Anderson and Robinson (2026) study climate-belief revisions following wildfires and global warming attention, and link it to cross-sectional differences in allocations to fossil-fuel-exclusion mutual funds within Sweden's premium pension system, measured about three and a half years after the wildfires. Instead, I compare active portfolio changes from pre-event allocations, tracing the timing and subsequent path of portfolio adjustments over shorter horizons around disasters.

[7] This pattern is consistent with Deryugina's (2013) finding that abnormally warm weather makes conservatives more likely to report that the effects of global warming have already begun.

2 Data and Measurement

This section describes the data (Section 2.1) and the climate interpretation measure constructed to capture the local information environment (Section 2.2). As outcomes, I use data on household, mutual fund manager, and other institutional investor portfolio holdings, which I combine with records of climate disasters and local newspaper coverage of each disaster to capture how disasters are interpreted locally.

2.1 Data

Households: local investment advisers. To represent household portfolio decisions, I use the portfolios reported by local U.S. registered investment advisers (RIAs) that predominantly serve household accounts. Advisers file Form ADV with the SEC, where they report their client composition, assets under management, and office locations.I combine quarterly Form ADV filings with raw Form 13F holdings to obtain the location and the clientele of each 13F filer. Following Pan et al. [2025], I define an adviser as serving predominantly household accounts when individuals and high-net-worth individuals make up a strict majority of its clients, and as local when all its offices are in one county.[8] The final sample contains 4,040 advisers in 490 counties, covering all 50 states, the District of Columbia, and Puerto Rico, from 2009 to 2024 (Table 1). Appendix Table A.6 shows that FactSet classifies most of these advisers as investment advisers (66.4 percent) or as private banking and wealth-management firms (30.6 percent).

Table 1: Registered Investment Adviser Sample Characteristics, 2009–2024

Statistic	All registered advisers	Household-serving	Household-serving, single-county
Registered investment advisers	8,489	4,439	4,040
Adviser-quarters	222,832	111,373	87,853
U.S. counties represented	1,439	850	490
States and U.S. jurisdictions represented	52	52	52
Median RAUM ($ millions)	593.8	468.0	426.1
Median client accounts	439	745	681
Median advisory offices	10.0	8.0	6.0
Advisers ever holding any ETF (%)	87.1	97.3	97.1
Advisers ever holding a green ETF (%)	36.3	45.6	44.0

Notes. This table reports characteristics of the registered investment advisers in the sample over 2009–2024. The universe comprises investment advisers with a direct Form ADV filing or identified as relying advisers on Schedule R; exempt reporting advisers are excluded. Advisers serving predominantly household accounts have a strict majority of individual clients. Single-county advisers report offices in one county. Regulatory assets under management, client accounts, and offices are measured at each adviser's latest observation. ETF indicators require a positive reported position. U.S. jurisdictions include the 50 states, the District of Columbia, and Puerto Rico.

Registered advisers held 41.1 percent of the $10.3 trillion U.S. ETF market at the end of 2024 and 45.2 percent of its green segment, against 20.3 and 17.9 percent for every other Form 13F institution combined.[9] ETFs use is concentrated exactly in the group of advisers serving households. Of the $38.2 billion of green ETFs held by all registered advisers at the end of 2024, advisers serving predominantly household

[8] Appendix A.7 reports the construction and selection details.

[9] Appendix Table C.1 shows ETF adoption, portfolio allocation, and market shares for registered advisers and for the household-serving group.

accounts held $18.8 billion, and local household advisers alone held $11.7 billion. The final sample of local household-serving RIAs holds 7.2 percent of the regulatory assets reported by all registered advisers, while holding 20.9 percent of their ETF dollars and 30.7 percent of their green ETF dollars. More than 97 percent of these advisers have ever held an ETF, and more than 44 percent have held a green ETF.

I aggregate 13F filers into stable adviser-level portfolio units,[10] assign each unit to a county, and aggregate positions to the county-quarter level. I restrict the sample to advisers that never change county, so that within-county portfolio changes are not mechanically driven by relocations.

I aggregate ETF holdings across local advisers to measure the allocation of the combined ETF wealth of the households they serve. Under the assumption that these households reside in their advisers' county, pooling holdings gives each household's ETF investment its corresponding weight in the county portfolio. Advisers serving more households or managing larger household investments therefore contribute more to the aggregate. The county green share is

$$g_{ct} = \frac{\sum_{a \in c} \sum_f V_{aft} G_f}{\sum_{a \in c} V_{at}^{ETF}} = \sum_{a \in c} \frac{V_{at}^{ETF}}{\sum_{b \in c} V_{bt}^{ETF}} g_{at}, \tag{1}$$

where G_f identifies green ETFs, V_{aft} denotes adviser a's holdings in ETF f, $V_{at}^{ETF} = \sum_f V_{aft}$, and g_{at} is the adviser's green ETF share.[11]

ESG labels influence investor demand beyond differences in underlying holdings [Baker et al., 2024]. I therefore define an ETF as green if its name in CRSP contains at least one of a list of climate and transition keywords, or if Morningstar assigns it one of a selected set of climate and clean-technology themes. I exclude sectors whose climate mandate is ambiguous[12] and funds with only a social mandate. The rule classifies 227 funds as green over 2009–2024, and Appendix A reports the full classification. Since green ETFs became common in adviser portfolios only around 2015, the main analysis covers 2015–2024 (Section 3).[13] At the end of 2024, 179 green ETFs were active, with a market value of $94.45 billion, or 0.92 percent of the U.S. ETF market.

To ensure that an adviser's county is also where its clients live, I require the principal office and every branch office to be in the same county, and I manually review the names of all retained advisers. I exclude internet-based advisers and advisers whose offices are all in one county but whose clientele is not local. To validate the local definition, I verify that the share of an adviser's stock portfolio invested in firms headquartered in its own state exceeds the share obtained by randomly reassigning adviser locations. I compute the green share within an adviser's ETF holdings, so that changes in mutual fund holdings reported on the same Form 13F do not enter the household measure.[14]

[10] The FactSet parent rollup and the S34 manager identity define the candidate aggregations. I split a parent into separate portfolios only when they belong to distinct advisers and are reported separately throughout. This keeps a portfolio's holdings unchanged when it is reported under a different filer identity. Appendix A.8 reports the criteria.

[11] Adviser-level regressions weighted by pre-disaster shares of county ETF holdings approximate this aggregate response and yield very close estimates, as reported in Appendix Table C.2. They capture changes in green shares within advisers but omit the contribution of changes in advisers' relative ETF holdings; Online Appendix OA.A derives the conditions for equivalence. Equal-adviser-weighted regressions instead measure the average adviser's response, giving advisers equal importance regardless of the number of households they serve or the ETF wealth they manage. Although this differs from my target, I report these estimates in the Appendix: the results remain statistically significant, while their magnitudes change with the different estimand.

[12] Metals and mining, resource extraction, upstream and midstream energy, commodities, agriculture and food, nuclear, water, and batteries.

[13] Green ETF market value grew from $2.0 billion in 2009 to a peak of $105.6 billion in 2021.

[14] Some advisers also manage mutual funds, and their 13F filings then include the funds' holdings. This overlap matters little for ETF positions. In the same quarter, mutual funds in S12 hold only 1.23 percent of ETF market value and 2.11 percent of green ETF

Finally, I investigate who the households behind these portfolios are. ETFs are held largely by households, directly or through their advisers. Advisers whose clients are mostly high-net-worth individuals held 46.9 percent of their reported 13F assets in ETFs by 2023, against 6.6 percent for pensions and endowments [Moussawi et al., 2025]. In 2024, 16.9 million U.S. households, 13 percent of the total, owned ETFs. These households are relatively wealthy, with a median income of $140,000 and median financial assets of $500,000 [Investment Company Institute, 2025], against a national median household income of $87,460 in 2025 [U.S. Census Bureau, 2026]. Since 58 percent of U.S. families hold equity directly or indirectly [Board of Governors of the Federal Reserve System, 2023], more than one in five investing households owns ETFs. These facts suggest that the local and household-account restrictions keep the part of an adviser's 13F portfolio that most likely reflects its clients' choices, especially for green ETFs.

Mutual fund managers. I combine fund holdings from the Thomson Reuters S12 database, fund characteristics and manager names from Morningstar Direct, manager profiles from Revelio Lab., and the filings of the managing firms with the SEC to build a panel of fund portfolios located where their managers work. Since mutual fund investors cannot be located, I study the portfolio managers who actively rebalance the funds, whose location I observe. I locate each fund at the offices of its sub-adviser, or of its adviser when it has none, from Form N-SAR until 2017 and Form N-CEN thereafter, following Alekseev et al. [forthcoming], and from Morningstar portfolio-manager records. This office often differs from the fund's headquarters (Section 4.3). I also match 1,161 funds to 2,348 named managers by combining Morningstar Direct and Revelio, which allows me to measure the managers' political views from their political donations. I construct fund-quarter holdings from S12 and compute changes only between consecutive quarterly reports. I also restrict the sample to actively managed U.S. equity funds.[15] Following Sammon and Shim [2023], I decompose these changes into a component implied by fund flows and an active rebalancing component. To measure the environmental profile of the stocks the funds trade, I use the Environmental Pillar score from LSEG [LSEG, 2024], the part of the ESG rating related to climate change. Appendix Figure A.4 maps the county-weighted geography of fund management.

I classify a mutual fund as having an ESG mandate if its name includes a sustainability term or if Morningstar classifies it as having an intentional ESG mandate. I exclude funds with an ESG mandate from the analysis of managers (Section 4), because their portfolios follow climate considerations and incentives unrelated to local disasters [Alekseev et al., forthcoming]. I retain Low Carbon-only and unlabelled funds in the manager sample. I also compare flows into ESG and confirmed non-ESG funds at the end of Section 3, using a classification that additionally includes Morningstar climate-theme funds and excludes Low Carbon-only and unlabelled funds.

Other institutional investors. The remaining 13F filers are mostly investment advisers and investment companies serving firms, municipalities, pension funds, and other institutions, together with hedge funds, banks, and insurers. I locate hedge fund managers, and other advisers whose clients or decisions appear local (e.g., those serving firms or municipalities), with the same rules as for the household advisers, applied to their

market value, and only 0.59 percent of S12 funds hold any green ETF. The Appendix shows that the results are similar for flows normalized by lagged assets and for dollar flows.

[15] I exclude funds whose Thomson Reuters objective is international, municipal bonds, bond and preferred, or balanced [Alok et al., 2020, Kacperczyk et al., 2008], passive funds and exchange-traded funds, identified by CRSP flags and fund names [Appel et al., 2016, Sammon and Shim, 2023, Alok et al., 2020], and funds with less than $5 million in assets or fewer than ten stocks [Alok et al., 2020].

Form ADV filings under full registration or as exempt reporting advisers. I map 1,561 hedge fund managers to 117 counties and 6,757 other registered advisers to 1,438 counties, counting every office. Following Baik et al. [2010], I place pension funds, banks, and insurers at the county of their headquarters, where their portfolios are managed centrally.

Climate disaster episodes. I build disaster data from the NOAA Storm Events Database, which records storms and other weather events at the county level. I group related county records into disaster episodes, so that a disaster that hits several counties counts as a single event while each county keeps its own exposure.[16] I assign each episode a dominant hazard type and measure its severity from inflation-adjusted property and crop losses, fatalities, and injuries. I combine these measures into an index of the episode's aggregate impact.[17] For each sample, I retain the episodes occurring in the period of analysis that are in the top 15 percent of this index, within or across hazard types, and exclude episodes that had another local episode in their pre-period. Appendix Table A.1 reports the characteristics of the episodes by hazard type, and Figures C.1 to C.2 map them by county.

Local news. To measure how disasters are discussed locally, I collect newspaper articles from ProQuest TDM Studio. I query articles whose text includes climate and global-warming vocabulary, as well as extreme-weather and disaster vocabulary, yielding 1,485,896 disaster articles from 527 outlets. Restricting to articles that cover a disaster over 2015–2024, the corpus holds 782,714 articles, of which 724,969 come from local outlets and 57,745 from national ones. I combine the articles with local newspaper circulation data from the Alliance for Audited Media and Editor & Publisher, complemented by device-measured readership from Comscore. The circulation data give ZIP-level newspaper market shares, which I use to aggregate outlet-level measures to the ZIP and county level. Using these weights, I construct a sample of 433 outlets reaching 2,134 counties. Appendix A provides additional details on the construction of the corpus and of the circulation weights.

Additional data. To construct the instrument for local interpretation (Section 5), I combine newspaper ownership data from [Ewing, 2026] with the ownership panels of the Northwestern Medill Local News Initiative. I date ownership changes to the quarter using SEC Form 8-K filings and reports of newspaper and company acquisitions. The remaining sources enter as controls, such as characteristics of ZIP codes, counties, advisers, mutual funds, institutional investors, and disasters, or serve to construct placebo events and other newsworthy events that coincide with disasters. Appendix A describes the source and construction of each of these datasets.

2.2 Local Information and Interpretation Environment

Climate beliefs are shaped by local discourse, through family and friends, news from different sources, and social media. Thirty-nine percent of Americans hear about global warming in the media at least once a month, and 17 percent hear people they know talk about it as often [Leiserowitz et al., 2026]. Local news reaches people through the same channels. Seventy-three percent of Americans get local news from friends,

[16] I audit every multi-event episode for coherence and split those that fail into sub-episodes, so that each episode approximates the footprint of one physical shock (Appendix A).

[17] The construction of this index builds on a Pareto approximation of the tail of disaster losses [Barro and Jin, 2011]. Appendix A provides additional details on the construction.

family, and neighbors, 64 percent from local television, half from local social media groups or forums, and a third directly from local daily newspapers, down from 43 percent in 2018 [Pew Research Center, 2024b,a]. I therefore measure how much local newspapers cover each disaster and how they interpret it. These two measures aim to capture the narratives shared locally and the local discussions they trigger.

Although only a third of residents read a local daily directly, newspapers shape much of what circulates locally. First, they originate most of the stories that set the agenda of other media,[18] and even small outlets can spark wider discussion [King et al., 2017]. Second, their articles reach readers beyond print, through websites, news aggregators, and social media.[19] Finally, local newspapers also reach investors. Retail and institutional investors rely on geographically proximate information [Engelberg and Parsons, 2011, Baik et al., 2010, Bernile et al., 2015], and local newspapers are "a key channel through which institutional investors acquire geographically scattered information" [Kang and Nam, 2025].

To construct a measure of local climate interpretation, I classify every paragraph of the disaster articles with a hierarchical teacher–student pipeline of large language models. I first define the classes and hand-label about 300 paragraphs, which I use to tune the prompts of a teacher model, GPT-5.5. The teacher then labels a training set, on which I train two local student models, DeBERTa and Qwen. DeBERTa classifies the full corpus and passes uncertain cases to Qwen. The classifier first detects whether a paragraph discusses a specific disaster event, identified in time and place. If it does, the classifier records whether the paragraph links the event to climate change and how strongly, and its stance on climate change. The stance can be affirming, denying, neutral, lukewarm (accepting global warming but minimizing its importance), or reporting contested opinions, and the classifier also records whether the stance is the reporter's own. In addition, I identify other common topics in each paragraph, the most frequent being damages and the human toll (damage to homes and victims). Against my own hand labels on held-out paragraphs, the two student models reach an F_1 score of 0.87 and 0.88 on the climate link, above general-purpose models such as gpt-4.1-mini (0.85) and gpt-5.4-mini (0.77).[20] Appendix Table A.4 shows two examples. The first, from the coverage of Hurricane Helene, links the storm to the burning of fossil fuels without ever mentioning "climate change," so a keyword search would miss it, while the classifier labels it as an affirming climate link. The second, from the coverage of Hurricane Katrina, tells the human toll through one person's story and makes no link to climate change.

The classifier yields two measures, the first one being local disaster coverage. Let o index outlets, c counties, and e disaster episodes, and let w_{co} be outlet o's circulation share in county c, normalized to sum to one within the county, and A_{oe} the number of articles on episode e that outlet o publishes in the four weeks from the episode's start. Local disaster coverage of episode e in county c is the circulation-weighted number of articles,

$$A_{ce} = \sum_{o} w_{co} A_{oe}. \tag{2}$$

[18] Pew Research Center, *How News Happens*, 2010, https://www.pewresearch.org/journalism/2010/01/11/how-news-happens/.

[19] Of the 546 local newspapers in the sample, at least 468 have an active website and 293 a Twitter account. Within a county, their print market shares track their web market shares (correlation 0.78; Appendix Figure D.1), and their Twitter audiences track their print circulation (rank correlation 0.84; Appendix Figure D.2). News aggregators matter most for the audiences of small and local outlets [Calzada and Gil, 2020, Athey et al., 2021].

[20] Appendix Figure A.1 shows the pipeline. Appendix Tables A.2 and A.3 report the performance of each model on every classification task, and Appendix A details the training and validation procedures. Appendix A also reports the full set of classes, including paragraphs on disasters in general, climate change, climate policy, and peripheral climate topics.

When coverage enters the regressions, I use $\log(1 + A_{ce})$ to allow for diminishing returns to additional coverage.

The second measure is local climate interpretation, the extent to which local coverage of a disaster interprets the event as a result of climate change. Let C_{oe} be the number of outlet o's articles on episode e that link the event to climate change, and $m_{oe} = C_{oe}/A_{oe}$ the outlet's interpretation ratio, defined for outlets with $A_{oe} > 0$. I aggregate these ratios to the county with the same circulation weights, renormalized over the outlets that cover the episode

$$M_{ce} = \frac{\sum_o w_{co}\, \mathbf{1}\{A_{oe} > 0\}\, m_{oe}}{\sum_o w_{co}\, \mathbf{1}\{A_{oe} > 0\}}. \tag{3}$$

M_{ce} is the share of climate-linked articles in the coverage of the average local outlet that covers the episode, with outlets weighted by their readership in the county. Each outlet counts in proportion to its readership, whatever the number of articles it publishes on the episode.[21] The measure averages 0.172 across county–episode observations, with a standard deviation of 0.157, so an increase of 10 percentage points corresponds to 0.64 standard deviations. In a county served by a single outlet, such an increase means one more climate-linked article in every ten articles on the episode. In a county served by two outlets with equal circulation, it means either one more climate-linked article in ten in both outlets, or two more in ten in one outlet and no change in the other.

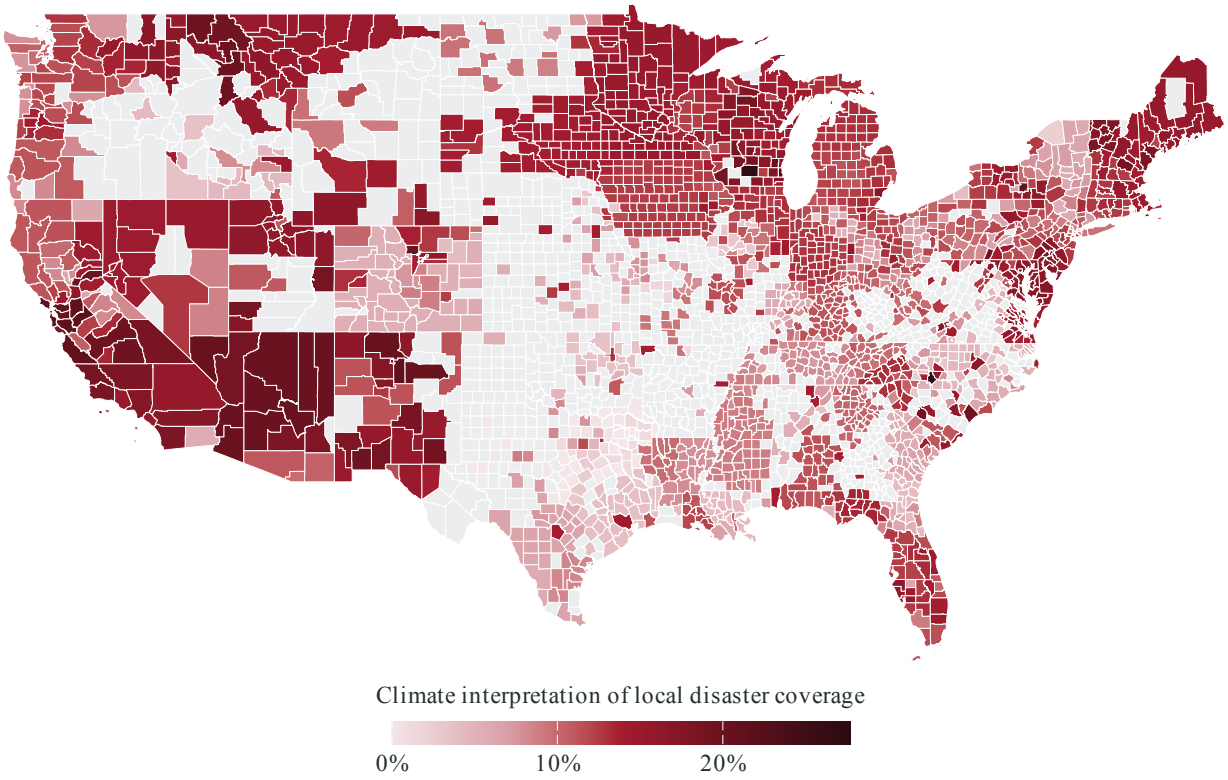
Figure 1 shows the average local climate interpretation of disaster coverage in each county. Interpretation varies widely across the country: it is highest along the West Coast and in the Southwest, and lowest in Texas and much of the South. It also rises over time, from 6.5 percent of a county's local disaster coverage before 2015 to 17.9 percent from 2015 on.

To separate the role of the outlet from that of the disaster, I decompose the variance of coverage and of climate interpretation across outlet–episode pairs into an outlet component, an episode component, and their match, as in the literature on matched employer–employee data [Abowd et al., 1999, Kline et al., 2020]. The outlet component captures stable differences across outlets, such as the political views and green attitudes of their audience and owner. The episode component captures the characteristics of the disaster, such as its year, hazard type, and damages. The match captures how a given outlet covers a given disaster beyond what the outlet and the disaster predict on their own. Appendix Figure D.3 shows that coverage depends mostly on the episode, which accounts for 55.4 percent of its variance against 19.5 percent for the outlet, mainly through the hazard type.

Climate interpretation instead lies mostly in the match, at 54.1 percent of its variance. The event component accounts for 40.7 percent of the variance in climate interpretation. Over 90 percent of its explained part comes from the year of the event, because the interpretation of each hazard type changes over time. Stable outlet and audience differences, together with common event characteristics, including time, still leave more than half of the variation unexplained. A substantial part of the variation therefore concerns how a specific outlet interprets a specific disaster, while serving a given audience at that time. Part of the match reflects how articles reach an outlet. Many local newspapers belong to chains, groups of newspapers under the same owner, and a chain often writes an article once and prints it in several of its newspapers. Among the outlet-event pair characteristics I observe, chain copies account for 92 percent of the explained part of the match. I leverage this variation to provide causal evidence on the effect of local interpretation in

[21] As a robustness check, I also use the pooled share $\sum_o w_{co} C_{oe} / \sum_o w_{co} A_{oe}$, which weights outlets by both their readership and the number of articles they publish on the episode.

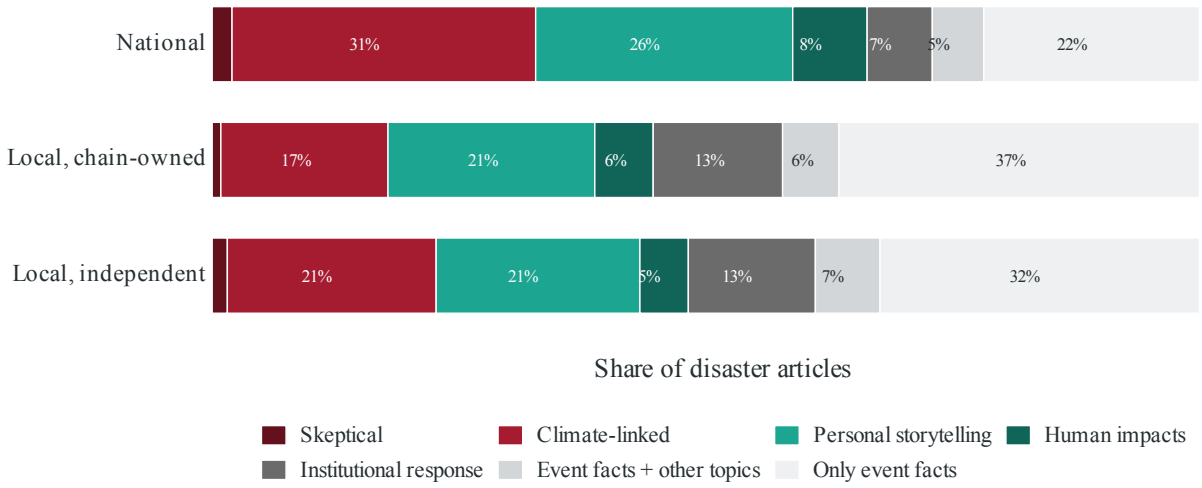
FIGURE 1: LOCAL CLIMATE INTERPRETATION OF DISASTER COVERAGE



Notes. Mean local climate interpretation of disaster coverage in each county, M_{ce} in Equation (3), over the disasters covered by its local outlets. Grey counties have no local coverage of disasters in the sample.

Section 5.

FIGURE 2: COMPOSITION OF DISASTER COVERAGE BY OUTLET OWNERSHIP



Notes. This figure plots the composition of disaster coverage for national outlets, chain-owned local outlets and independent local outlets. Shares are calculated among 424,888 disaster articles published from 2015 through 2024: 41,016 national, 353,748 chain-owned local and 30,124 independent local. A local outlet is chain-owned when its owner holds two or more newspaper titles in the quarter of publication, and independent when its owner holds one. Categories are mutually exclusive. Skeptical treatment takes precedence over climate linkage, and climate linkage over the narrative and contextual categories. An article is assigned to Only event facts when every relevant block reports the specific event. It is assigned to Event facts + other topics when at least one relevant block is not about that event. This block either discusses hazards as a class, a trend or a projection, or covers green products, sustainability or climate regulation.

The way a disaster is represented in the news therefore depends on the type of outlet that covers it. Figure 2 shows the composition of disaster coverage for national outlets, chain-owned local outlets, and independent local outlets. The key difference is in the share of disaster articles linked to climate change, which is 30.8 percent for national outlets, 16.9 percent for chain-owned local outlets, and 21.1 percent for independent local outlets. Across all disaster articles, chain-owned local outlets link disasters to climate change 4.2 percentage points less often than independent ones, and the gap remains statistically significant within the same quarter (Appendix Table D.1).

I interpret this measure as an equilibrium local interpretive environment: it captures whether local discussions and narratives, proxied by newspaper coverage and framing, map a realized disaster into climate meaning. Section 5 will isolate the effect of one source of exogenous supply-driven variation in this measure coming from shocks in newspaper ownership structures to leverage the variation in local interpretation described above.

3 Households and Fund Investors: Green ETFs and ESG Fund Flows

I find that a higher share of local news coverage explicitly linking the occurring extreme weather event to climate change is associated with a significantly larger increase in the green-ETF share of household portfolios over the following two quarters. To show this, I begin with a difference-in-differences design that compares affected counties with clean comparison counties around the same episode and, among affected

counties of the same hazard in the same year, counties whose newspapers read the disaster differently. The estimate is stable across all five columns of Table 2, conditional on detailed controls for the county's politics, climate beliefs, demographics and climate attention before the event, and with county fixed effects. I then present and rule out the alternative explanations by progressively saturating the specification, and finally introduce two alternative strategies that address the two remaining concerns: comparing advisers within the same event, and separating the reading of the current event from the county's belief predisposition.

More climate-interpreted coverage moves the local portfolio toward green funds over the two quarters after the disaster, on both the extensive and the intensive margin; in counties without significant climate-linked coverage, disaster coverage moves portfolio turnover without a corresponding net tilt toward sustainable funds.

3.1 Empirical Strategy

My objective is to estimate how the local interpretation of a disaster affects the green share of household portfolios. The ideal experiment would compare the response of investors in otherwise identical counties, with the same level of exposure to the same disaster at the same time, and differing only in how the local discourse (proxied by newspapers) has interpreted the disaster. I approximate it by exploiting two sources of variation. First, because the timing of a disaster is exogenous, comparing the change in the green share of affected counties with that of unaffected counties around the same episode identifies the response to the disaster at average attribution. Second, among the affected counties, I compare those with similar observable characteristics, exposed to the same type of disaster in the same year, differing in the climate framing share of their local newspapers. In practice, I implement this comparison by estimating the following local-projection regression at each horizon h from the extreme weather event onset:

$$\Delta_h q_{ie} = \gamma_h M_{ie} + \theta_h D_{ie} \cdot M_{ie} + D_{ie} \cdot \delta^h_{k(e),y(e)} + X'_{it}\left(\Gamma_h + D_{ie} \cdot \Lambda_h\right) + \alpha^h_{s(i)} + \eta^h_e + \varepsilon_{ieh}. \tag{4}$$

The unit i is a county or an adviser, observed around episode e starting in quarter $t(e)$, at horizon h. Treatment is repeated and non-absorbing: a county can be affected by several episodes over the sample, and the comparison counties of one episode can be the affected counties of another. Therefore, I construct the sample of treatment and control units around each episode with a clean-window design following Dube et al. [2025]: treated units are those in the disaster zone of the episode with no earlier local disaster in the two quarters before the start, and control units are clean comparison counties[22] not affected by that episode or by any other major disaster within a window.[23]

The outcome is the change in the share of a county's exchange-traded fund holdings in environmentally focused funds, q_{it}, measured through the household-serving local advisers of Section 2 and aggregated to the county of the adviser's office. As described in Section 2, holdings are pooled across the county's advisers before taking the share, and adviser-level regressions are weighted by pre-disaster shares of county ETF

[22] More precisely, clean comparisons are defined relative to each episode and each horizon: for horizon h, control county-quarters have no local disaster in the stabilization window before the reference quarter and none between it and $t+h$. Treated county-episodes have no other local disaster through horizon h; episodes starting in the same quarter are kept, each county–episode weighted by one over the number of episodes of its county in that quarter.

[23] Following Dube et al. [2025], the first comparison is a local-projection difference-in-differences with clean controls. Because episode fixed effects absorb everything common to an episode, the effect of D_{ie} is estimated within each episode: the regression is a stacked version of their quarter-level event study estimator in which each episode forms its own stack, as in Cengiz et al. [2019]. It identifies the response to the episode under parallel trends. Results are robust to adding a spatial condition on the clean controls, removing comparison counties close to the disaster.

holdings. I define the cumulative change from the quarter before the onset, the cumulative portfolio-weight change, as $\Delta_h q_{ie} \equiv q_{i,t+h} - q_{i,t-1}$. To isolate active rebalancing from valuation drift, I construct the portfolio weight change at fixed pre-event price: for each episode starting in quarter t and each horizon h, I rebuild the county portfolio using quantities observed at $t+h$, valuing every ETF at the pre-event price vector $P_{j,t-1}$,

$$q_{i,t+h|t-1} = \frac{\sum_j G_j P_{j,t-1} Q_{ij,t+h}}{\sum_j P_{j,t-1} Q_{ij,t+h}}, \tag{5}$$

and $\Delta_h q^{FP}_{ie} \equiv q_{i,t+h|t-1} - q_{i,t-1|t-1}$.[24] The pre-event-price measure isolates active reallocation, while the total measure captures the net observed green exposure. When both are positive and close to each other in magnitude, I can exclude both that advisers' observed increase in relative exposure is the artifact of simple revaluation, and that active re-allocation has been simply the result of a counterbalancing response to negative drift to maintain desired exposure, therefore identifying an active change into a new desired allocation. Taking the change from the quarter before the episode removes the level of each county. The difference removes any level difference between the two groups. The main text reports the two-quarter cumulative pre-event-price portfolio change, alongside the current-price change, rescaled to basis points.[25]

D_{ie} is an indicator function and takes value 1 when i is in the disaster zone of episode e, while M_{ie} is the climate-interpreted share of local coverage of the episode.[26] The coefficients of interest are θ_h, the differential effect of interpretation in affected relative to unaffected units, and $\gamma_h + \theta_h$, the total interpretation effect among affected units. $\delta^h_{k(e),y(e)}$ are affected × hazard × year fixed effects, where $k(e)$ and $y(e)$ are the hazard type and year of episode e, capturing the average response of affected counties in each hazard type and year; $\alpha^h_{s(i)}$ are state fixed effects, where $s(i)$ is the state of unit i; and η^h_e episode fixed effects, absorbing the level of each disaster. The episode fixed effects compare units in the disaster zone with the unaffected units of the same disaster. The affected × hazard × year fixed effects compare the interpretation effect among affected units within the same year and the same hazard type.

X_{it} is a vector of county controls that may be correlated with both local interpretation and the portfolio response. Each control enters both alone (Γ_h) and interacted with the affected indicator (Λ_h). It always includes disaster coverage, among both affected and unaffected units, to control for news attention to the disaster, with additional controls depending on the analysis.

The key additional controls, which hold fixed what could make a county both receive a climate reading of the disaster and buy green assets in any case, are the county's Democratic vote share at the last presidential election and the share of its residents worried about global warming; its income, college share, population and density; the size of its advised portfolio in the previous quarter; the share of its local coverage devoted to climate policy and green topics in the quarter before the episode; and its own damage, deaths and injuries. Section 3.3 discusses the specific confounders each set of controls rules out, as well as robustness checks and alternative approaches to identify the interpretation effect. Standard errors are clustered by county.

3.2 Results

Interpretation Effect on Green Household Demand. The full results are presented in Table 2. I find that a one-standard-deviation increase in local climate interpretation of the disaster raises the pre-event-price green

[24] This follows the same logic as Calvet et al. [2009], adapted to fit the local projection estimation framework.

[25] Results are robust to non-cumulative outcomes, alternative post-event windows, and alternative outcome definitions.

[26] M_{ie} is missing for county-episodes with no local coverage, so the interaction is estimated on covered county-episodes.

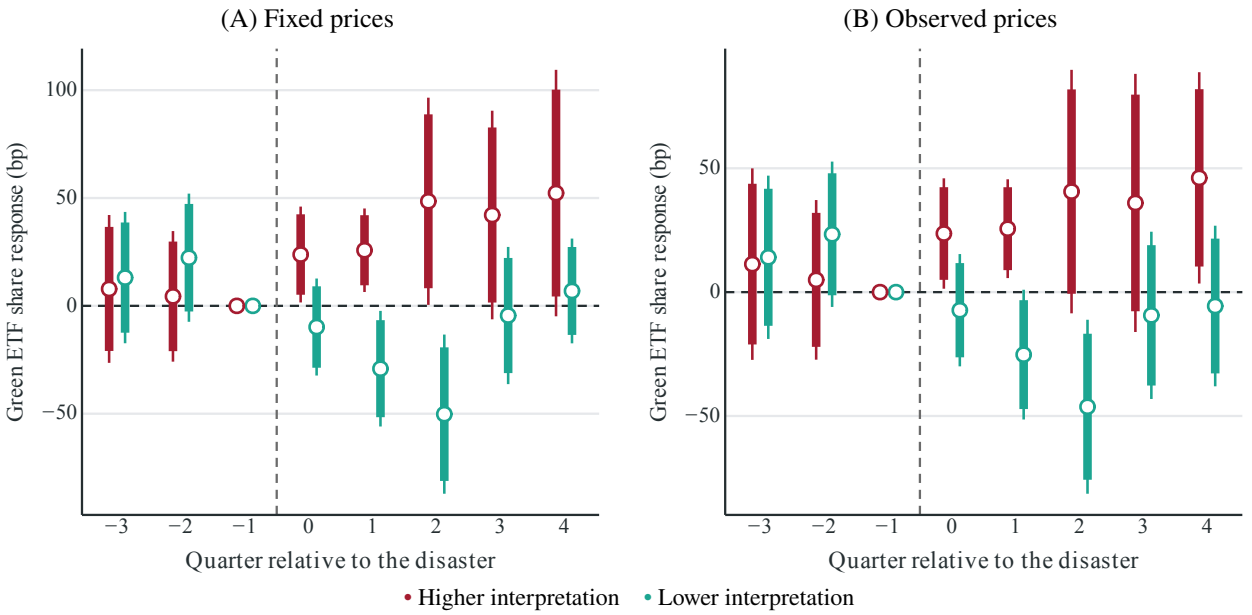
Table 2: County green-ETF portfolio response

	Dependent variable:									
	Green-ETF share response (basis points)									
	Pre-event prices					Current prices				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Affected × Interpretation	25.48**	25.86**	22.46*	30.31**	22.76*	25.87**	25.58**	22.53*	29.50**	22.55*
	(10.13)	(12.33)	(12.18)	(12.58)	(12.44)	(10.46)	(12.50)	(12.25)	(12.62)	(12.28)
Affected × Attention	.86	4.91	3.64	1.69	−9.36	.46	4.55	3.40	1.79	−9.16
	(10.93)	(10.92)	(10.30)	(10.06)	(10.01)	(11.17)	(10.84)	(10.24)	(10.08)	(10.11)
Total Interpretation effect among affected counties	26.70**	27.08**	23.47*	31.32**	22.73*	27.02**	26.73**	23.43*	30.43**	22.54*
	(10.45)	(12.44)	(12.26)	(12.53)	(12.31)	(10.75)	(12.61)	(12.32)	(12.58)	(12.16)
Observations	105,662	105,662	105,662	105,662	105,662	105,662	105,662	105,662	105,662	105,662
R^2	.028	.029	.031	.032	.102	.030	.030	.033	.033	.106
Affected county-episodes	244	244	244	244	244	244	244	244	244	244
Episode and quarter FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
State FE	✓	✓	✓	✓		✓	✓	✓	✓	
County FE					✓					✓
Affected × hazard × year FE		✓	✓	✓	✓		✓	✓	✓	✓
Financial and demographic controls			✓	✓	✓			✓	✓	✓
Own-county damage, deaths and injuries			✓	✓	✓			✓	✓	✓
Political and belief controls				✓	✓				✓	✓
Lagged climate attention				✓	✓				✓	✓
Affected × controls				✓	✓				✓	✓

Notes. This table reports estimates of the county green-ETF share response to disasters. The outcome is the average county green-ETF share over the quarter of the episode and the two following quarters minus its average over the two quarters before, in basis points, at pre-event prices, held at their level in the quarter before the episode (columns 1–5), and at current prices (columns 6–10). The sample covers high-salience flood, hurricane and coastal episodes, 2015–2024. Affected is one for counties in the episode footprint. Interpretation is the climate-interpreted share of local coverage of the episode and Attention the log volume of that coverage, both over the first four weeks of the episode and standardized; one standard deviation of Interpretation is 15.7 percentage points. Each regression also includes Interpretation and Attention and, in columns 1 and 6, Affected; their coefficients are not reported. The total Interpretation effect among affected counties is the sum of the Interpretation coefficient and its Affected interaction, with its standard error from their joint covariance. Financial controls are lagged log portfolio assets; demographic controls are income, college share, white share, population and population density; political and belief controls are the prior Democratic vote share and the county share worried about global warming; lagged climate attention is the share of local coverage on climate policy and green topics in the quarter before the episode. Standard errors, in parentheses, are clustered by county. A nonparametric bootstrap that resamples counties with replacement (999 draws) gives $p = .036$ for Affected × Interpretation in column 4 and $p = .032$ in column 9. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

share of affected counties by 30 basis points relative to the unaffected counties linked to the episode.[27] The estimate for observed prices is within one basis point of the pre-event-price estimate, therefore proving that the increase is not confounded by price drift in either direction.

FIGURE 3: DISASTER RESPONSE OF COUNTY GREEN ETF SHARES BY LOCAL INTERPRETATION, BY QUARTER



Notes. Conditional disaster responses of the county green ETF share for counties with local interpretation above and below the median, by quarter relative to the flood, evaluated at a common reference. Changes are relative to quarter −1, normalized to zero. Calendar-quarter and disaster-episode FE and Hit × Year terms. Controls include lagged assets, prior politics, demographics, framing and attention. Thin and thick bars are 95 and 90 percent confidence intervals, with two-way clustering by county and disaster episode. Joint test of the High–Low gap in quarters −4 to −2: $p = 0.463$ (A), 0.397 (B).

Of two counties linked to the same disaster, the one whose local coverage connects the event to climate change ten points more often, thirty percent of articles in place of twenty, holds 19 more basis points of green share at pre-event prices two quarters later. That corresponds to $0.66 interpretation-induced extra millions reallocated to green ETFs by the median county-quarter ETF holdings of these advisers ($340.3 million), and $5.04 millions at the mean holdings ($2.61 billion). The two-quarters cumulative change induced by 10 percentage point share belongs to the 82nd percentile of all absolute two-quarter changes on this support.

Results are robust to different specifications of this design.[28]

Extensive and Intensive Margins. Per standard deviation of interpretation in affected counties, the share of advisers in the county holding a green ETF rises 5.76 percentage points, and 5.94 among advisers reporting from the same county across the two quarters, so the increase comes from advisers already there. Over the same two quarters that share rises 1.35 percentage points on average across the county-episodes of this sample, so the interpretation response is more than four times the ordinary post-disaster movement. Among

[27] The estimate is column 4 of Table 2, with all controls and their interactions with D_{ie}. Standard errors are clustered by county, the more conservative choice: $p = .017$, against $p = .009$ when clustering by episode. With county fixed effects the estimate is 23 ($p = .068$); with state-by-year fixed effects, 20 ($p = .213$).

[28] For example, robust to extensions of the non-overlapping window to concurrent-episode weighting functions (see Appendix B for their constructions). Moreover, Appendix Table E.1 reports the same control columns with interpretation measured over the calendar quarter, and Table E.3 reports the same estimation at the adviser level. The comparison does not rest on the control counties: comparable differential when estimating the slope among the affected counties alone, while the same contrast among unaffected counties is indistinguishable from zero (Appendix E, Figure E.5).

advisers in the disaster zone, the response is 19 basis points per ten percentage points of interpretation, which is around $224 thousand at the median adviser ETF book and $446 thousand at the mean.[29]

Extension to ESG Mutual-Fund Flows. The main analysis uses geographically detailed ETF holdings of household-serving advisers to relate local climate interpretation to portfolio allocation. To establish external validity beyond green ETFs, I extend the analysis to ESG mutual-fund flows.[30] Disaster exposure is assigned at the fund's headquarters state, assuming that headquarters location proxies for clientele exposure, motivated by evidence of geographic familiarity in mutual-fund selection [Bailey et al., 2011]. Using this approximation, Marshall et al. [2021] find higher net inflows into green mutual funds relative to matched nongreen funds when disasters occur in the funds' headquarters states. In Figure 4 I compare relative net ESG inflows, measured against net inflows into non-ESG funds, across disasters with high and low climate interpretation, holding disaster attention and other covariates constant.

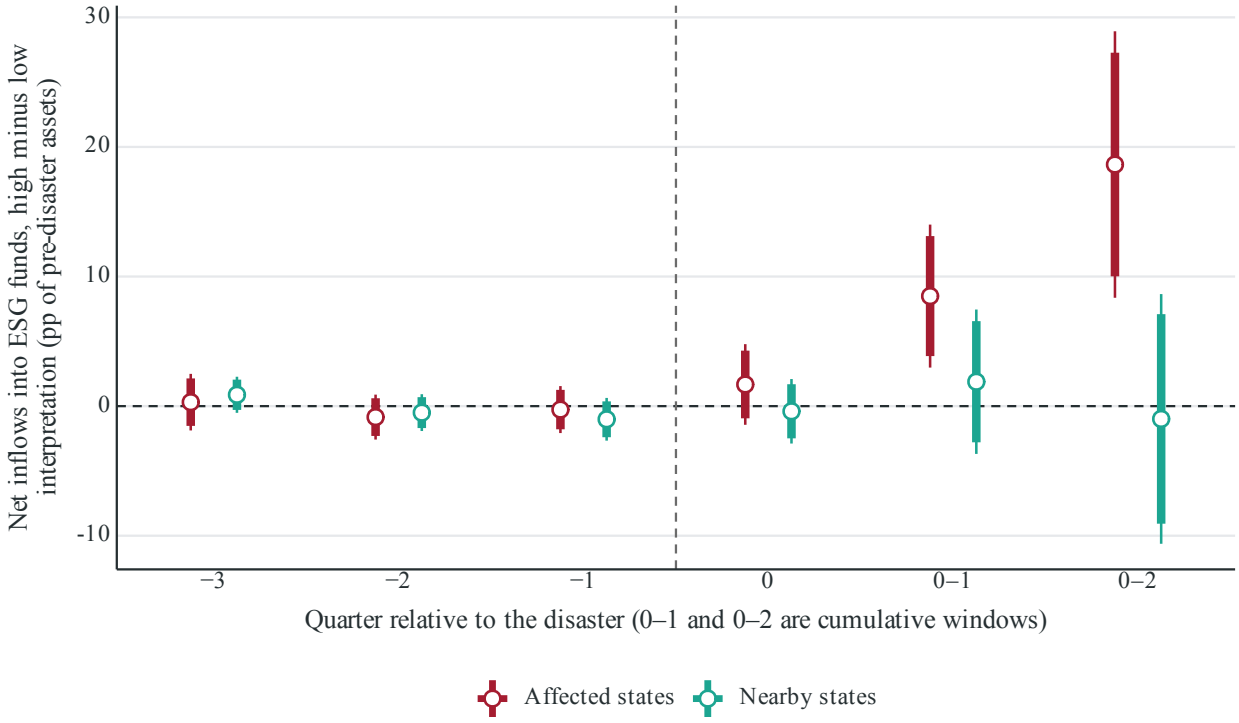
Fund fixed effects absorb persistent differences in fund flows. ESG-status-by-quarter fixed effects absorb broad national trends in ESG investment by allowing ESG and non-ESG fund flows to vary separately each quarter. Hazard-by-exposure-group fixed effects account for average differences across disaster types and geographic exposure.

Among funds headquartered in disaster-affected states, relative net ESG inflows are larger following high-interpretation disasters, by 8.49 percentage points of pre-disaster assets over two quarters and 18.64 points over three quarters. The corresponding differences in neighboring unaffected states are statistically insignificant. These findings provide suggestive evidence that the interpretation–investment relationship extends from advisers' green ETF allocations to ESG mutual-fund flows.

[29] See Table E.4.

[30] ESG funds are identified by a dated green or sustainable fund-name classification, a Morningstar intentional-ESG designation effective before the disaster, or a Morningstar climate-theme flag. The comparison group comprises confirmed non-ESG mutual funds; ETFs, low-carbon-only funds, and funds with ambiguous classifications are excluded.

FIGURE 4: CLIMATE INTERPRETATION AND FLOWS INTO ESG MUTUAL FUNDS



Notes. This figure plots the High-minus-Low difference in relative net inflows into ESG mutual funds, measured against non-ESG funds, separately for funds headquartered in disaster-affected and neighboring unaffected states. High and Low interpretation are defined using the median interpretation among disaster-affected state observations, with the same cutoff applied to both groups. The outcome is net dollar flow divided by pre-disaster assets, in percentage points. Pre-event points use individual quarterly flows; post-event points accumulate flows from Q0, with the same asset denominator. Each horizon is estimated separately on fund–disaster observations, 2015–2024. High interpretation and its interaction with ESG status enter separately by band, together with ESG status × Affected. Attention and its ESG interaction also enter separately by band. Fixed effects are fund, ESG status × event-calendar quarter, and hazard × band. Controls are log disaster damage and its ESG-status/band interactions, and log pre-disaster assets. Thin and thick bars are 95% and 90% confidence intervals, clustered by disaster episode.

3.3 Identification

Equation (4) nests two designs. The first identifies the response to the disaster, through the experience of the event, its salience and the attention it receives, by comparing affected units with the clean comparison counties of the same episode. The response is therefore identified under no anticipation and parallel trends between affected and comparison counties, similarly to the designs that compare units by their exposure to the same disaster [Dessaint and Matray, 2017, Alok et al., 2020, Gallagher, 2014], with clean controls that correct the bias of two-way fixed-effects estimators when disasters are repeated and non-absorbing [Dube et al., 2025].[31] The second identifies the interpretation effect from the affected counties: it compares affected counties exposed to the same type of disaster in the same year, with similar observable characteristics and damage, whose local interpretation of the disaster differs [similarly to Djourelova et al., 2024, who compare

[31] No anticipation: the green share of a county does not move before an episode in response to it. Parallel trends: absent the episode, and given the controls, the green share of affected and comparison counties would have followed the same path; with repeated disasters, this holds for each episode a county is linked to [Dube et al., 2025]. The clean window of two quarters keeps earlier local episodes out of the pre-period of affected counties and any change in exposure out of the comparison counties through the horizon.

respondents exposed to the same disaster by the climate coverage of their local newspaper]. The unaffected counties are still used to estimate the episode fixed effects, as well as to include more saturated fixed effects, including county and state fixed effects, the controls and γ_h. Therefore, θ_h comes from a triple difference.[32]

The interpretation measure already captures the reading of the event that forms locally, whether the county's outlets supply it or cater to their readers' predisposition. The estimates measure the response to local interpretation from either source. A bias arises when the readers who demand climate interpretation of a disaster would have bought green assets after it without that interpretation. The identifying assumption is that, conditional on these controls, the climate reading of a county's coverage of a disaster is unrelated to how its green share would have changed under a different reading: affected counties discussing the disaster more as climate change would otherwise have changed their green share similarly to those discussing it less as climate change; Figure E.5 shows the event-study path with the coefficients on leads and lags, enabling visual inspection of potential pre-trends to assess the plausibility of this identifying assumption.

By exploiting progressively more fine-grained variation, the columns of Table 2 restrict the level at which a confound would have to operate. The episode fixed effect absorbs the disaster itself, its size, its national coverage, everything common to the affected and unaffected counties of each specific episode, including anything else in its quarter of onset. Because each horizon is estimated separately, the episode fixed effect also absorbs common shocks in the following quarters, such as national events or news. Furthermore, the state fixed effects absorb the average change in the green share of the counties in each state. On top of this, column 2 controls for the average response of affected counties in each hazard type and year, so that the interpretation effect does not come from comparing fundamentally different disasters, or from underlying trends in climate salience or market trends in sustainable products picked up by comparing disasters in different years.

Column 3 adds financial and demographic controls, alone and interacted with the affected indicator, including the size of the advised portfolio in the previous quarter, in case richer counties respond more, and the county's own damage, deaths and injuries to check that the interpretation effect is not driven by how strongly a specific county was affected by the event.[33] Column 4 adds the county's prior Democratic vote share, its share worried about global warming and its climate attention in the quarter before onset, so that the reading of this disaster is net of how much its newspapers already write about climate, and their interactions with the affected indicator, so that the interpretation effect is estimated net of the disaster response of greener, more Democratic and more worried counties.

Column 5 replaces state with county fixed effects. On the stacked panel a county appears in every episode it is linked to, almost always as a nearby or distant comparison county,[34] so the county effect absorbs its trend in the green share and its usual change around large disasters, wherever they occur; the interpretation effect is then read from how an affected county's change after its own disaster departs from that usual change, compared with the other affected counties in the same hazard and year. The interpretation effect does not fall with county fixed effects: if the estimate only reflected counties that react strongly to every disaster, it

[32]The within-episode design of Section 3.4.2 and the newspaper-ownership design of Section 5 restrict the sample to the advisers or counties affected by or close to a disaster. Djourelova et al. [2024] similarly restrict their sample to respondents exposed to a local disaster, comparing those surveyed just before and just after it.

[33]Damage could itself be the confounder: a county more strongly impacted may be less likely to buy green or may be forced to liquidate assets, and local news in a severely affected county may turn to fatalities, losses and emergency information, at the expense of climate interpretation articles, so that both lower interpretation and lower rebalancing toward green appear in the most affected counties.

[34]Even a county that is affected by some episode is affected in only 0.4 percent of the episodes it is linked to, and is a comparison county in the rest.

would have vanished once each county is compared with its own usual response. Nor is interpretation a stable trait of local outlets: a fixed outlet component explains only 5.5 percent of its variance, and most of it reflects how a specific outlet reads a specific disaster (Section 2.2). This suggests that the effect is tied to the interpretation of the current event.

Remaining confounds must operate within a county and among affected counties of the same hazard type and year. For example, an affected county whose newspapers read a flood as climate change would have to raise its green share around the event, by more than the other counties affected by similarly impactful floods that year, for reasons other than interpretation-driven climate concern, and those reasons would have to be correlated with how its newspapers read the flood.

Some identification concerns remain, which Section 3.4 addresses specifically: that the interpretation effect picks up predetermined county beliefs, attitudes or trends unrelated to disasters, which I address with placebo tests and with the newspaper-ownership design of Section 5, where the interpretation that local outlets supply shifts with ownership changes; that it reflects comparisons across disasters, which the within-episode design of Section 3.4.2 replaces with a comparison of advisers at the same distance from the same disaster in the same state; and that it mostly captures the salience of the disaster, which I address with ancillary analyses that apply the same design to other outcomes. It also discusses threats to reading the response as a demand shock for sustainability.

3.4 Alternative Explanations

I address here the possible alternative explanations that would threaten either the reading of my previous results as the effect of the interpretation of the specific event, or the interpretation of the observed response as evidence of revealed sustainability demand shock.

3.4.1 Threats to Interpretation

Interpretation or Generic Climate Salience. One possibility is that interpretation effect is simply capturing general contemporaneous or diffused climate talk or attention unrelated to the event. Table F.1 addresses that concern directly by adding my corresponding local measures of same-quarter and lagged non-disaster-specific climate attention to the analysis. I included these terms both as controls and as heterogeneities, allowing generic non-disaster climate salience (e.g., due to climate policy events occurring in the same quarter) to compete against disaster interpretation in mediating the portfolio response. The climate talk interactions are all indistinguishable from zero, while the interpretation coefficient, estimated against these competing moderators, remains positive at 29bps in the most demanding case (against a baseline result of 30bps).

Interpretation of Current Event or Predetermined County Type or Predisposition. While I ruled out that a higher volume of disaster coverage is driving the results, a remaining concern is that the coefficient on climate change contextualization may be driven by counties where climate is already more salient, more discussed, or more behaviorally relevant and, therefore, may be picking up pre-existing local concern about climate change and not how disaster experiences are interpreted.

One possible explanation of the response is that it is driven by attitudes the county already held, with the disaster as the occasion on which they are acted upon. If residents more concerned about climate change buy green assets whenever a salient local event occurs, and if coverage in those same counties is also more climate-interpreted, this would imply a spurious correlation between interpretation and green reallocation.

Table F.1 suggests that this is not the case. Each row adds one county characteristic, interacted with the disaster-zone indicator, with climate interpretation, and with their product; the four rows here use the county-level share of individuals worried about global warming, the perceived scientific consensus, the county Democratic vote share, and the share of population that has college degree. All four triple interactions are statistically indistinguishable from zero, and the interpretation effect evaluated at the mean of each remains between 37 and 45 basis points, significant at the 1% level. Measured climate attitudes have little explanatory power for the magnitude of the interpretation response. The controls, measured before the event, hold fixed the readers' climate concern in levels, so a climate reading of the coverage may still go with a change in that concern around the event. Moreover, if local climate interpretation mainly captured a general propensity of climate-concerned counties to buy green assets after any salient local shock, the same pattern would appear after earthquakes, which no coverage reads as climate change; there is no differential green tilt after earthquakes in counties with more climate interpretation (Appendix Table F.2). Finally, Section 5 uses changes in the ownership of the county's newspapers, which shift the climate-interpreted share of their disaster coverage, to isolate the interpretation that local outlets supply.

Interpretation or Attention. Interpretation could also proxy for attention to the disaster: more heavily affected places may be both covered more and interpreted more, so that the interpretation effect would be the disaster's own attention effect showing through a correlated measure. Affected counties are hit by the disaster, so they most likely pay attention to it in any case, and the specification controls for its damage. and every specification controls for coverage. To exclude that counties with a stronger climate reading simply move their portfolios more after a disaster, I examine how gross turnover[35] in affected and unaffected counties changes with local disaster coverage and with interpretation (Table F.3). Turnover of the whole reported book rises with coverage in both affected and unaffected counties, by 0.107 and 0.081 log points per standard deviation, while its response to interpretation in affected counties is null. High-interpretation counties do not move their portfolios systematically more than low-interpretation ones. The differential portfolio response to interpretation therefore provides suggestive evidence validating that attention and conditional interpretation (here captured by disaster experience, coverage and its newspaper interpretation) are separate independent constructs.[36] The newspaper-ownership design of Section 5, which shifts the interpretation that local outlets supply while holding coverage fixed, separates the two further. Section 6 separates them in household insurance demand: after wildfires, which do not make flood risk salient, flood-insurance take-up still rises with climate interpretation.

3.4.2 Threats to Event Link and Specificity

Comparable or Same Event. A concern with the county estimates is how much the interpretation effect captures a response to similar events against the variation in interpretation within one episode.[37] The fixed effects and controls of Section 3.1 compute the interpretation effect only across disasters of the same type, in the same year, with damage held fixed. Such disasters may still differ in the national attention and the

[35]Following Calvet et al. [2009], turnover is the value of absolute quantity changes over the quarter at start-of-quarter prices, divided by the starting reported book value, aggregated to the county and measured as log(1 + turnover in basis points).

[36]The event-study plots dividing counties at the median of interpretation (Figure 3) show that low-interpretation counties tend to reduce their green share in the quarters after the disaster, suggesting, if anything, that they are reacting by accommodating the demand.

[37]NOAA splits one storm system into separate episodes at state and forecast-office boundaries and by hazard type, so many episodes have a single affected county with observed advisers, and a separate intercept for the affected counties of each episode is too demanding at the county level.

national interpretation they receive, in the state where they occur and its politics and policies, and in the local economy of the affected counties, and these can correlate with the local interpretation. The within-episode design removes these differences: it compares advisers at the same distance from the same disaster in the same state, so that the event's size and national coverage, the state's politics and policies and the distance to the damage are the same for the advisers compared. To compare portfolio responses to the same disaster, I use advisers in affected counties and nearby advisers within 50 miles of the affected area, and I measure local interpretation at the adviser's office ZIP code. I estimate

$$\Delta_h q_{ie} = \theta_h M_{ie} + \alpha^h_{e,s(i)} + \lambda^h_{e,b(i,e)} + \mu^h_{c(i)} + X'_{ie}\gamma_h + u_{ie,h}, \qquad h \in \{0, 1\}, \tag{6}$$

where $\Delta_h q_{ie}$ is the change in green-ETF share from the preceding quarter to the disaster quarter or the following quarter, measured in basis points at pre-event prices. M_{ie} measures interpretation in ten-percentage-point units, with a common coefficient across affected and nearby advisers. X_{ie} includes the controls and their group interactions described in the Online Appendix.

I compare portfolio changes among advisers facing the same disaster in the same state whose local coverage interprets it differently. Episode by state fixed effects absorb the disaster itself, with its footprint, impact and national coverage, and everything common to the state, such as differential trends in political or climate views, in the insurance market and in state policy. Episode by band fixed effects allow affected advisers and nearby advisers 0–25 and 25–50 miles from the affected area to have different average responses to each disaster. With county fixed effects, the comparison is relative to the county's average portfolio response across episodes in which its advisers are affected or nearby.

The identifying assumption is that, conditional on the controls, advisers in the same state at the same distance from the same disaster would otherwise have changed their green share alike. For example, an adviser whose newspapers read a flood as climate change would have to raise its clients' green share by more than advisers at the same distance from the same flood in the same state, for reasons other than interpretation-driven climate concern that correlate with how its newspapers read the flood.

A ten-percentage-point increase in interpretation is associated with a 7-basis-point increase in green-ETF share in the disaster quarter and a 12-basis-point increase by the following quarter, both relative to the preceding quarter (Table F.4, column 3). Standard errors are clustered by disaster episode and ZIP.

3.4.3 Threats to Revealed Demand Shock

Household Demand Shocks or Advisers. I restrict the sample to local advisers serving predominantly household accounts. Still, most assets are managed on a discretionary basis, so portfolio changes can reflect both client preferences and advisers' own views. I therefore compare county voting, which proxies for clients' political views, with advisers' political views measured from staff donations, including both in the regression. If advisers' own views drive the response, their political leanings should help explain it. The response is stronger in less Democratic counties but does not vary significantly with advisers' own political views (Table F.5). This supports a household-demand interpretation, consistent with Pan et al. [2025], who find that county political preferences predict household-serving advisers' portfolio choices even after accounting for advisers' own political leanings.

The effect is largest where the two parties are near even, and it is also positive and significant in Republican-majority counties, while it is not statistically distinguishable from zero in strongly Democratic counties. This pattern is consistent with the survey evidence, although the differences across political groups

are not jointly significant (Appendix Table F.5).

Less Democratic advisers are less likely to hold any green ETF to begin with. However, conditional on holding a green ETF, their own views do not significantly affect the disaster response, and, more importantly, the interpretation effect[38]. Taken together, this evidence suggests that the green ETF response to the local interpretation records the beliefs of the households the advisers serve, and not the advisers' own.

Within-ETF Reallocation or Liquidity Shocks. One concern might be that the effect could be purely mechanical: disasters may generate liquidity needs, withdrawals, or precautionary adjustments that change ETF shares even without any active move toward green assets. In particular, advisers may cut ETFs first to meet their liquidity needs because ETFs tend to be particularly liquid and easy to trade, and may be held as a liquidity buffer. If among ETFs, non-green ETFs are liquidated even more aggressively, then the ETF denominator can fall and green ETF shares can rise even without a genuine tilt toward green funds. Instead, I find that green ETF assets rise alongside non-green and total ETF assets, while total observed 13F holdings show no statistically significant change (Table OA.F.1). ETF and whole-portfolio flows show no systematic contraction in counties and episodes with higher climate interpretation (Table OA.F.2). Moreover, the interpretation effect holds also among advisers in nearby counties not directly affected by the extreme weather event. These findings do not support liquidation as the explanation for the increase in the green ETF share.

Conflict of Interest. A separate concern is that advisers encourage clients to buy their green products, or to buy green ETFs and then accommodate that demand by selling the underlying stocks. About 2% of the local advisers in my sample sponsor or manage the ETFs in their holdings. I find that when advisers purchase green ETFs, they also tend to purchase the underlying constituent stocks directly (Table OA.F.3). This pattern does not support systematic offsetting sales within the observed portfolios. Advisers may also use a disaster as an occasion to rebalance client portfolios and justify their fees. I control for advisers' prior ETF and whole-portfolio turnover. The interpretation response does not vary significantly with a one-standard-deviation increase in either measure (Table OA.F.4).

4 Mutual-Fund Managers and Active Rebalancing

The second setting I study is actively managed mutual funds. In this context, I can observe the investment teams that make the decisions and measure the political views of the managers themselves, as well as the local information environment they are embedded in. This allows me to measure political views more closely, to separate the decisions of delegated portfolio managers, and to study the effect among professional institutional investors. I find that a higher share of local news coverage explicitly linking the occurring extreme weather event to climate change is associated with a meaningfully larger shift of the portfolios of nearby mutual fund managers toward stocks of environmentally-friendly firms in the quarter of the event.

[38] Only smaller advisers show an effect of the firm's own leaning on the average disaster response. Among the green ETF holders, more Republican staff means a smaller adjustment after the disaster, by 12 basis points per standard deviation of the firm's lean. The effect sits among the firms with small ETF books, the bottom third, with a median book of eleven million dollars, about two percent of their regulatory assets, and is absent when advisers are weighted by their share of the county's book. They still show none on the interpretation response.

4.1 Empirical Strategy

The hypothesis is that managers are influenced by the narratives surrounding a disaster occurred near to where they work, and it affects their portfolio decisions, particularly toward more sustainable assets when local coverage interprets the disaster as climate change. To show this, I compare, across funds with offices within 50 miles of the affected area whose local newspapers read the event differently, the shift toward stocks of more relative to less environmentally friendly firms in the quarter of the event. I study how the portfolios of these funds change around climate-related disasters from 2015 to 2024. In each quarter, I link a fund to the nearest disaster and to every disaster that hits one of its offices.[39] For fund f and episode e, I estimate the following regression

$$\begin{aligned}\Delta_h q_{fe} = \gamma_h D^d_{fe} M^d_{fe} + \sum_{b\in\{a,w,n\}} \theta^b_h D^b_{fe} \cdot M^b_{fe} + \sum_{b\in\{a,w\}} D^b_{fe} \cdot \delta^{b,h}_e + D^n_{fe}\,\delta^{n,h} \\ + X'_{fe}\Gamma_h + \sum_b D^b_{fe} \cdot Z^{b\prime}_{fe}\Lambda^b_h + \mu^h_{c(f,e)} + \kappa^h_{s(f)} + \eta^h_e + \varepsilon_{feh},\end{aligned} \tag{7}$$

where M is the share of local coverage that interprets the disaster as climate change (Equation 3). The fund's offices fall into four distance bands. D^a_{fe} is the share of the fund's offices in the counties hit by the disaster, D^w_{fe} the share within 100 miles of them, D^n_{fe} the share within 50 miles, a subset of the previous band, and D^d_{fe} the share more than 100 miles away. M^b_{fe} averages M over the fund's offices in band b. γ_h is the effect of interpretation among managers more than 100 miles away. The coefficient of interest is the total effect among nearby managers, $\theta^w_h + \theta^n_h$.

The stocks that mutual funds trade carry no green label like the green ETFs of Section 3, so I measure each stock's environmental profile with its Environmental Pillar score (Section 2.1). The outcome, $\Delta_h q_{fe}$, is the green tilt of the fund's trading, the change in its weight in stocks in the top tercile of the score minus the change in its weight in stocks in the bottom tercile, over the quarter of the disaster, in basis points of the fund's scored stock portfolio. As for households (Equation 5), I compute the weights at the beginning and at the end of the quarter using prices at the beginning of the quarter. The outcome therefore captures the managers' own trades and is not driven mechanically by changes in prices. I also report the change at current prices. To validate the measure, I use the approach and data of Alekseev et al. [forthcoming], extended to 2023, which treat local heat waves as shocks to managers' climate concerns. A fund with all its offices exposed to such a heat wave raises its green tilt by 65 basis points at beginning-of-quarter prices (Appendix Table G.1).

I then place each fund's offices in distance bands around the disaster, with equal weight on each office. Managers are affected when an office lies in a county hit by the disaster and nearby when it lies within 50 miles of one. Managers more than 100 miles away are the comparison group. A fund with offices in several bands belongs to each in proportion to its offices there. I exclude a fund from the analysis of a disaster when it has an office within 100 miles whose county had no local coverage of the disaster, since local interpretation is not defined there. I also exclude it when stocks with an Environmental Pillar score make up less than half of its stock portfolio before the disaster.[40]

[39] As disasters I consider floods, hurricanes, wildfires, severe storms, heat waves, and droughts. I exclude a fund from the analysis of a disaster when one of its offices was hit by another disaster in the same or the previous quarter.

[40] This leads to exclude just around 2.7 percent of observations. For the median fund in the final sample, 94 percent of the stock portfolio before the disaster is scored.

Finally, instead of focusing on directly hit counties, where the disaster caused the most damage,[41] I estimate the effect on managers located in counties near those directly affected.

Isolating the effect of local interpretation on managers has two main challenges. First, how local outlets interpret a disaster may be endogenous and reflect stable characteristics of the place and of the managers who live there, and not the event itself. Namely, counties whose outlets often write about climate change, or with a given political leaning, may also host managers who respond differently to any disaster. Second, the disaster itself can change the stocks the managers hold and the flows into their funds, whatever its coverage. To address the first challenge, I include fixed effects for the county from which the fund is managed, which absorb how funds managed from that county respond to disasters in general. To address the second challenge, I include disaster fixed effects to absorb the movements of greener and browner stocks in the quarter of the disaster, which are common to all managers.

η_e^h are disaster fixed effects that absorb market-wide movements of greener and browner stocks in the quarter of the disaster. $\delta_e^{b,h}$ are band-by-disaster fixed effects for affected managers and those within 100 miles, capturing each group's average response to each disaster, including its damage. $\delta^{n,h}$ is a level common to all disasters for managers within 50 miles. $\mu^h_{c(f,e)}$ are fixed effects for the county from which the fund is managed, which absorb how funds managed from that county respond to the large disasters in the sample. For a fund with offices in several counties, this is the county with most of its offices, and the one closest to the disaster in case of a tie. $\kappa^h_{s(f)}$ are fund-style fixed effects, which remove average differences across investment styles. Columns 4 and 8 also include fund fixed effects. Each fund's change in tilt is then measured against its own average.

The effect of interpretation is therefore identified by comparing managers at the same distance from the same disaster whose local coverage interprets it more or less as climate change. As for households, I estimate the interpretation effect among the units exposed to the disaster and compare their response with that of a comparison group. In this application, I focus on the nearby managers as the exposed units, while the comparison group comprises managers more than 100 miles away.

X_{fe} contains the lagged size, age, and turnover of the fund. Z^b_{fe} contains, for the fund's offices in band b, the volume of local coverage and, depending on the column, county and political controls, as well as damage for the offices in the counties hit.[42] I also control for how the county's outlets interpret the other large disasters they cover, so that the comparison holds fixed the county's usual reading of climate events. I cluster standard errors by disaster.

4.2 Results

Table 3 reports the results. Panel A shows that nearby managers tilt their portfolios toward greener stocks when local coverage interprets the disaster as climate change. A ten percentage point higher interpretation share raises their green tilt by 69 basis points at beginning-of-quarter prices and by 71 basis points at current prices in the quarter of the disaster (Columns 2 and 6).[43] The estimates are similar with political controls

[41] Managers in counties directly hit by the disaster also face its direct consequences, such as damage and outflows from local investors.

[42] Political controls are the county's Democratic two-party vote share and the managers' political score, from the political donations of the named portfolio managers or, without them, of the management company. Appendix Table G.2 shows that the results are robust to including funds whose managers and management companies have no matched donations.

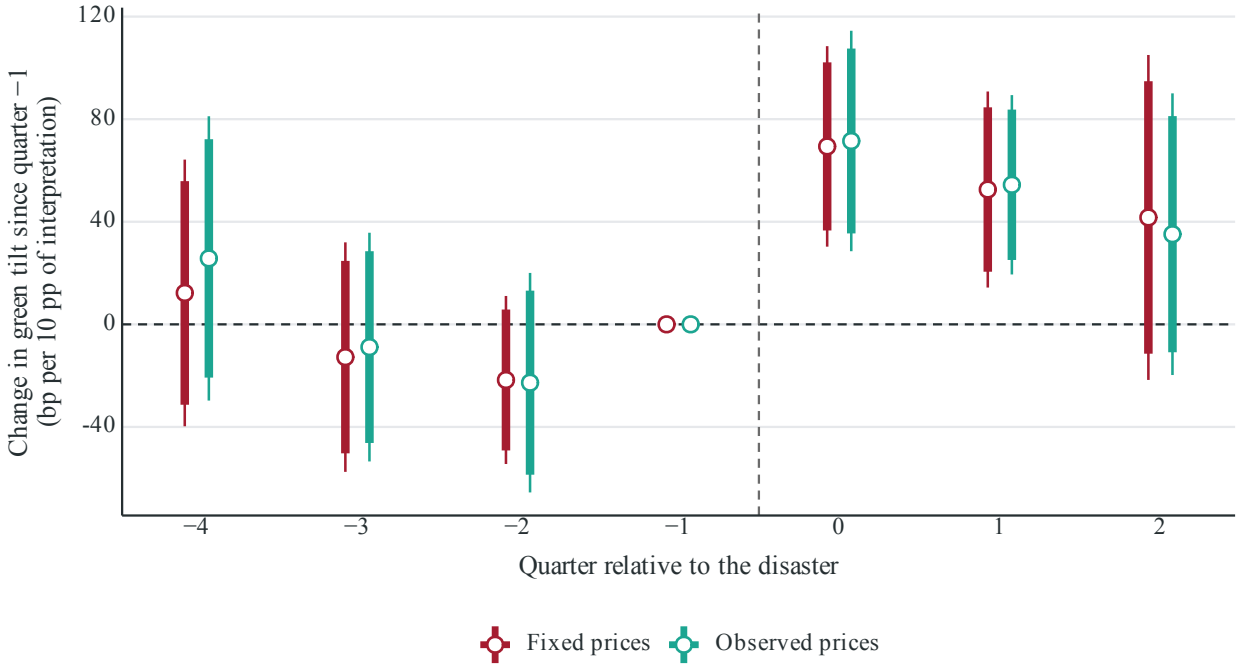
[43] The results are robust to restricting the sample to funds with all their offices in one county, to measuring each stock's environmental profile with its continuous Environmental Pillar score, and to placing the fund at all its offices or only at the office closest to the disaster (Appendix Table G.3).

TABLE 3: MUTUAL FUND PORTFOLIO RESPONSE

	Dependent variable: response in basis points							
	Pre-event prices				Current prices			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A. Fund level: change in the fund's Environmental-Pillar tilt								
Interpretation	69.34***	69.38***	66.25***	67.58***	71.03***	71.49***	68.74***	65.75***
	(20.11)	(19.87)	(18.99)	(19.86)	(20.85)	(21.85)	(21.08)	(21.56)
Attention	−2.05	−.85	−1.09	−1.95	−4.52	−3.79	−4.01	−4.43
	(2.84)	(3.12)	(3.09)	(3.36)	(2.98)	(3.36)	(3.28)	(3.58)
Observations (fund-episodes)	37,047	37,047	37,047	36,964	37,047	37,047	37,047	36,964
Nearby fund-episodes	1,712	1,712	1,712	1,710	1,712	1,712	1,712	1,710
R^2	.031	.032	.033	.093	.044	.045	.045	.102
Exposure × episode FE	✓	✓	✓	✓	✓	✓	✓	✓
County FE	✓	✓	✓	✓	✓	✓	✓	✓
Fund style FE	✓	✓	✓	✓	✓	✓	✓	✓
Fund FE				✓				✓
Fund controls	✓	✓	✓	✓	✓	✓	✓	✓
Damage and coverage controls	✓	✓	✓	✓	✓	✓	✓	✓
Exposure × county controls		✓	✓	✓		✓	✓	✓
Exposure × political controls			✓	✓			✓	✓
Panel B. Stock level: change in the fund's weight in a stock, High-E minus Low-E								
Interpretation		3.57***	3.56***			3.66***	3.71***	
		(1.28)	(1.28)			(1.33)	(1.29)	
Attention		.04	.03			−.09	−.13	
		(.27)	(.28)			(.29)	(.30)	
Observations (fund-episode-stocks)		5,426,327	5,426,327			5,426,327	5,426,327	
Fund-episodes		53,028	53,028			53,028	53,028	
R^2		.577	.577			.586	.586	
Fund × episode FE		✓	✓			✓	✓	
Fund style × stock × episode FE		✓	✓			✓	✓	
Exposure × episode FE		✓	✓			✓	✓	
Damage and coverage controls		✓	✓			✓	✓	
Exposure × political controls			✓				✓	

Notes. This table reports estimates of equation (7) on the fund-episodes with the political measures. In Panel A the dependent variable is the change in the fund's Environmental-Pillar tilt over the episode quarter, winsorized at the 1st and 99th percentiles; in Panel B it is the change in the fund's weight in the stock, High-E minus Low-E. Rows report the effects among managers within 50 miles of the episode. Interpretation is per 10 percentage points and Attention per 10 log points. Each regression also includes Interpretation and Attention among affected managers and among managers more than 100 miles away; their coefficients are not reported. Fund controls are lagged log total net assets, log age and turnover; county controls are the county's interpretation of its other episodes, log deaths, log population, log population density, log income, college share, unemployment rate and urban share; political controls are the county Democratic two-party vote share and the managers' political score. Exposure is the distance band of the fund's offices: affected, within 100 miles and more than 100 miles away; county and political controls enter separately for each. Standard errors, in parentheses, are clustered by episode. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE 5: INTERPRETATION EFFECT AMONG NEARBY MANAGERS, BY QUARTER



Notes. Effect of local climate interpretation on the green tilt of nearby managers by quarter relative to the disaster, per 10 percentage points. The outcome is the change in the tilt since the end of quarter −1, in the specification of Table 3, Columns 2 and 6. Thin and thick bars are 95 and 90 percent confidence intervals, with standard errors clustered at the disaster level.

(Columns 3 and 7) and with fund fixed effects (Columns 4 and 8).[44] In contrast, the volume of local coverage does not move the tilt, with coefficient on attention being small and never significant.

A one standard deviation increase in the interpretation share (16.4 percentage points) raises the tilt by 1.1 percentage points, corresponding to 2 percent of the average tilt of nearby funds before the disaster (56 percentage points). It also closes about a tenth of the gap in the tilt between a conventional fund and a fund with an ESG mandate, and it is 1.7 times the response to a local heat wave that causes injuries or deaths (Appendix Table G.1). In dollar terms, the average nearby fund, with $4.3 billion in assets, moves about $49 million from bottom-tercile to top-tercile stocks. Summed over the 29 nearby funds of the average disaster, which manage $125 billion, the shift reaches about $1.4 billion, or $110 million for the median disaster.

Figure 5 plots the effect quarter by quarter. In the three quarters before the disaster, the estimates are close to zero and jointly not significant. The effect is 69 basis points in the quarter of the disaster and persists while decaying for the next two quarters.

The effect is concentrated near the disaster. Among managers more than 100 miles away, a ten percentage point higher interpretation share raises the tilt by 4 basis points, about a twentieth of the effect among nearby managers (Appendix Table G.5).

A disaster can move the fundamentals and prices of the firms it affects, and greener and browner firms need not be affected alike. A tilt measured at the level of the fund can then mix the managers' choices with changes in the stocks they hold. In Panel B, I estimate the regression for each fund, stock, and disaster, with the change in the fund's weight in the stock as the outcome. At this level, I can include fixed effects for

[44]The estimates are smaller but remain significant when I replace the band-by-disaster and county fixed effects with fixed effects for each combination of distance band, hazard and calendar quarter (Appendix Table G.2, Columns 5 and 10).

each stock around each disaster, separately for each fund style. They account for everything that happens to the stock around the disaster and is common to the funds of that style that hold it, such as its return and fundamentals, the location of the firm and its own exposure to the disaster, national coverage of the firm and the shock to its industry. Fixed effects for each fund around each disaster absorb everything common to the fund's stocks. These include the attention and interpretation of its local coverage, its distance from the disaster, the damage in its county, its flows and its average trading, as well as the fund and county controls of Panel A. Interpretation, attention, damage and the band-by-disaster fixed effects therefore enter interacted with indicators for the top and bottom terciles of the Environmental Pillar score, as do the political controls in Columns 3 and 7. The Interpretation row reports the effect on top-tercile minus bottom-tercile stocks.

A ten percentage point higher interpretation share raises the weight of nearby managers in top-tercile relative to bottom-tercile stocks by 4 basis points at both beginning-of-quarter and current prices, and the estimates are similar with political controls (Columns 3 and 7). This estimate compares how managers whose local coverage interprets the same disaster more or less as climate change reweight the same stocks, relative to the other stocks in their portfolios and to funds of the same style that hold the same stocks.

4.3 Alternative Explanations

The effect could capture fund flows and not the managers' own choices. Funds managed in the counties hit by the disaster can face outflows from local investors [Tubaldi, 2021], and these outflows raise two concerns. First, outflows can vary with interpretation for reasons unrelated to the managers' views. For instance, climate narratives may be less common in counties where the same disaster was more destructive, so that interpretation picks up differences in outflows. Second, the sales that outflows force follow their own pattern across stocks [Coval and Stafford, 2007], which can create a tilt even after adjusting for the rebalancing due to flows.

To address the first concern, I measure trades net of the proportional scaling of positions with flows. A ten percentage point higher interpretation share raises the net active purchases of nearby managers in top-tercile relative to bottom-tercile stocks by 30 basis points of lagged assets (Appendix Table G.6, Column 2), suggesting that tilt comes from the managers' own trade. To address the second, I estimate the effect on managers outside the counties hit, and I keep the affected managers in a band of their own. Nearby managers do not face the outflows of local investors in the counties hit, so forced sales cannot explain their tilt. Appendix Table G.5 reports the results for affected managers.

Alternatively, the effect could capture the sale of firms located in the disaster area, if managers expect these firms to perform worse or want to limit their investors' exposure, and these firms happen to be browner. To test this, I split the tilt of Table 3, Columns 2 and 6, into the weight moved across distance bands of the firms and the choice among firms at the same distance from the disaster (Appendix Table G.7). The component across bands is small and not significant, while the component within bands is almost identical to the main estimates. The effect therefore comes entirely from the choice among firms at the same distance from the disaster. At the stock level, the effect for firms outside the disaster area is the same as the baseline estimate (Appendix Table G.8). Within the disaster area, managers shift toward the greener firms even more, by 19 basis points. Managers thus do not sell local firms as a block, but reweight toward the greener ones.

Finally, the effect could reflect the funds' investors and not their managers. Works on mutual funds and local disasters typically place a fund at its headquarters and treat that address as the location of its managers [e.g., Fich and Xu, 2025]. The headquarters is also a natural proxy for where the fund's investors live, because individual investors favor locally managed funds [Bailey et al., 2011]. In my data, the two locations

often differ. Among the 76,652 quarterly observations of funds with a managing office in Forms N-SAR and N-CEN or in Morningstar records of portfolio managers, the office is in the county of the headquarters in 65 percent of cases and more than 100 miles away in 30 percent. For sub-advised funds, the share more than 100 miles away is 63 percent.

Table G.9 uses this separation on the funds whose headquarters is known, with the specification of Table 3, Columns 1 and 2. In one regression, I estimate the effect for nearby managers whose headquarters is within 100 miles of the disaster, for nearby managers whose headquarters is farther away, and for funds whose headquarters is within 50 miles but whose managers are more than 100 miles away, using the interpretation of the headquarters county. The results show that nearby managers respond as much when their headquarters is far as when it is near. Funds whose headquarters are near the disaster but whose managers are far respond less, but not significantly, suggesting that response depends more on manager location than on headquarters location.

Taken together, this evidence points towards the tilt being more likely driven by the managers' response to the local interpretation of the disaster compared to explanations involving fund flows, sale of firms in the disaster area, or pressures by the funds' investors.

5 Separating Interpretation Narratives from Prior Beliefs and Preferences

The previous household, adviser and fund-manager designs compare units whose local coverage gives different interpretations to the same type of disaster, holding fixed the climate and political views of the county as well as those of advisers and fund managers. Event-study leads, placebo tests and increasing saturations ruled out confounders such as trend in green investment and a general propensity of climate-concerned counties to buy green assets, even after salient but non-climate shocks. As a result, the interpretation effect estimated so far already captures the effect of the local reading of a disaster, whether it stems from the narrative that the county's outlets supply at the time of the event or simply reflects the predisposition and prior beliefs of their readers, to which the outlets cater.

Here, I further investigate the role of locally supplied interpretation models and narratives [Schwartzstein and Sunderam, 2021] on top of prior beliefs and preferences, by testing whether the local interpretation moves portfolios when it is driven by changes in the narrative that the outlets supply, independently of the demand or preferences of their readers. I do so by adapting an instrumental variable approach that uses media ownership as a shifter of content [Martin and McCrain, 2019, Garz and Rickardsson, 2023, Ananyev and Volkova, 2024, Ewens et al., 2022, Kostovetsky et al., 2026]. In particular, I instrument the interpretation of the disaster in local news with the size of the chain that owned the county's newspapers, the number of newspapers in the sample held by the same owner. Larger newspaper chains are more detached from the counties they serve and cater less to local demand, supplying centrally produced copy.[45]

In practice, the instrument exploits changes in the ownership of the newspapers serving a county between two disasters of the same type. For each county, I compare the disasters of the same type that affected it directly or occurred within 50 miles, in particular those that occurred before with those that occurred after the chain size of its newspapers changed.

[45]Outside the United States, newspapers held by one owner share that owner's slant [Garz and Rickardsson, 2023], and acquisitions shift slant in a direction that depends on the acquirer [Ananyev and Volkova, 2024]; in American local broadcasting, ownership changes shift content [Martin and McCrain, 2019].

Specification and estimation. The sample contains the counties directly affected or within 50 miles and with positive local coverage of the event. I define the chain size of newspaper o in calendar quarter t, chain_size_{ot}, as the number of newspapers in the sample held by the outlet's owner as recorded prior to the event, so that an independent newspaper has a chain size of one. The county instrument is the mean of the logarithm of chain size[46] across the chained newspapers serving the county, weighted by their circulation, with ownership as recorded prior to the disaster quarter.

$$Z_{c,t(e)} = \frac{\sum_{o \in C_{c,t(e)}} w^D_{oc} \log \text{chain_size}_{o,t(e)}}{\sum_{o \in C_{c,t(e)}} w^D_{oc}}, \tag{8}$$

where C_{ct} is the set of chained newspapers serving county c under ownership as recorded prior to the disaster quarter and w^D_{oc} is the fixed circulation weight. The first-stage equation is

$$M_{ce} = \tilde{\pi} Z_{c,t(e)} + \tilde{\beta}_A A_{ce} + X'_{ce}\tilde{\gamma} + \tilde{\mu}_c + \tilde{\tau}_{t(e)} + \tilde{\lambda}_{h(e)\times b(c,e)} + u_{ce}, \tag{9}$$

where $Z_{c,t(e)}$ is the county instrument of equation (8), A_{ce} is local disaster coverage in logs, X_{ce} collects the logs of one plus own-county deaths and damage, the county Democratic two-party vote share at the previous presidential election and, in columns 2 and 4 of Table 4, the audience overlap of the county's newspapers with the event footprint,[47] $\tilde{\mu}_c$ are county fixed effects, $\tilde{\tau}_{t(e)}$ calendar-quarter fixed effects, and $\tilde{\lambda}_{h\times b}$ distance-band by hazard-family fixed effects; the comparison is among counties at the same exposure to disasters of one hazard family. I estimate

$$Y_{ue} = \beta_M M_{ce} + \beta_A A_{ce} + X'_{ce}\gamma + \mu_c + \tau_{t(e)} + \lambda_{h(e)\times b(c,e)} + \varepsilon_{ue}, \tag{10}$$

where u is a county or an adviser in county c and M_{ce} is instrumented with $Z_{c,t(e)}$. Adviser regressions weight each adviser by its share of the county-episode ETF book in the quarter before the episode. Standard errors are clustered by county.

Results. Table 4 reports two-stage least squares estimates of the instrumental variable design. Panel A reports the first stage at the outlet and county level, Panel B the county response of Table 2, Panel C the adviser response of Table C.2 and Panel D the mutual fund manager response of Table 3. Columns 1 and 2 use pre-event prices and columns 3 and 4 current prices; all columns control for coverage volume, own losses and the county's Democratic vote share, and columns 2 and 4 add the audience overlap of the county's newspapers with the disaster footprint. The coefficients remain stable in magnitude and significance across columns. As expected, the use of 2SLS reduces statistical power, but the weak-instrument-robust confidence sets exclude zero and the instrumented response is of the same magnitude as the local-projection estimates

[46] I use the natural logarithm to measure proportional changes in chain size, since the same change in the number of newspapers is more meaningful and impactful for a small chain than for a large one.

[47] Separating attention from interpretation requires holding fixed the incentive of the county's newspapers to cover the event. Beyond disaster coverage itself, I control for newspaper-audience overlap with the event footprint outside the county, inspired by the newspaper-market congruence designs of Snyder and Strömberg [2010] and Myers [2026]. Consider three nearby counties, A, B, and C. A disaster hits only A. People in B mostly read newspaper X, which has 90 percent of its readers in A. People in C mostly read newspaper Y, which has no readers in A. X therefore has more incentive than Y to cover the disaster, so people in B should receive more coverage than people in C, even though neither county is hit. Instrumenting coverage jointly with this measure gives similar interpretation estimates from a weakly identified system; that version is in the Online Appendix.

of Table 2 for counties and advisers.[48]

For mutual fund managers whose offices are all within 50 miles of the disaster, a 10-percentage-point increase in the supplied climate interpretation narrative raises their green tilt by 177 basis points at pre-event prices, about two and a half times the ordinary least squares estimate of Section 4 (69 basis points, Table 3).

Relevance and exclusion. The climate-interpreted share of disaster coverage differs persistently across owners, and newspaper chain size predicts that share at the outlet level; under a larger prior owner, local disaster coverage has a lower climate-interpreted share (Table H.1). Chain size is measured at the newspaper–quarter level; Panel A of Table 4 shows that its relation to the interpretation share extends to the event-specific share of the county's coverage.

The identifying assumption is that, conditional on the fixed effects, coverage volume and own losses, a change in the chain size of a county's newspapers is unrelated to any change in how its investors respond to the disaster, other than through the climate interpretation of the coverage they receive. Calendar-quarter fixed effects absorb everything common to the counties exposed in the same quarter, such as a rise in chain size or a fall in green shares across the country. Hazard-by-exposure-band fixed effects restrict the comparison to counties exposed to the same hazard type in the same exposure band: affected counties are compared with affected counties and nearby counties with nearby counties. The level of chain size could itself be endogenous: chains may own the newspapers of markets with particular economic, readership or political characteristics, and these could also drive the response to disasters. County fixed effects absorb these permanent differences between counties with larger and smaller chains, such as market size, politics, climate beliefs and the usual reading of their newspapers.

What remains is the county's log chain size at each disaster minus its average over the county's disasters.[49] Two kinds of ownership shock account for most of these changes. The first is an upstream shock, a change at the level of the owner: the owner expands or contracts through a merger or an acquisition by or of other owners, or through purchases and sales of other newspapers, as in the 2019 merger of Gannett and GateHouse.[50] The second is a direct change in ownership, in which the county's newspaper itself passes from one owner to another, as in the 2023 sale of the San Diego Union-Tribune by NantMedia to Alden Global Capital. Upstream shocks are the most common. Owners take these decisions at the chain level, for reasons that follow from the economics of the newspaper industry, such as declining advertising revenue, and are plausibly unrelated to the climate concern of the county's readers.[51]

Strictly prior ownership prevents the coverage of the disaster from determining the instrument; it does not make ownership random. Changes in chain size do not predict the county's latest presidential vote share, its climate beliefs measured before the sample, its portfolio change before the event, or the losses, distance and coverage of the disaster (Table H.2).

Disasters that attract greater attention may also lead newspapers to discuss their climate implications

[48] The interpretation share is a classifier output and carries measurement error that can bias its coefficient [Ash and Hansen, 2023, Hausman, 2001]; since ownership changes are unrelated to the classifier's errors on a given article, the instrument removes this bias as well.

[49] For example, the Hartford Courant, which holds 97 percent of the circulation of Hartford County, Connecticut, was owned by Tribune Publishing, a chain of 18 newspapers, at the time of the flood of 28 June 2020 and by Alden Global Capital, a chain of 56, at the time of the flood of 1 September 2021, both within 50 miles of the county.

[50] Table OA.G.1 in the Online Appendix analyses this merger on its own: after it, the interpretation share and the green-ETF response of affected and nearby counties are both lower where the county's pre-merger Gannett circulation share is higher.

[51] Acquisitions and consolidation in the industry follow declining advertising revenue, the liquidity needs of legacy owners, and cost cutting in roll-ups [Ewens et al., 2022, Gao et al., 2020].

more extensively, so the estimated interpretation effect could reflect how much attention the event receives. A change in ownership may also change how much the county's newspapers cover the disaster, so I instrument interpretation while controlling for the volume of local coverage. At the outlet level, chain size moves the interpretation share and leaves disaster coverage unmoved (Table H.1). A causal interpretation requires the ownership instrument to affect portfolios through interpretation, without also shifting unmeasured attention that independently affects investment.

6 Mechanisms

6.1 Salience

The pattern observed in green portfolio reallocation and fund flows and in mutual fund manager active rebalancing is consistent with salience and experiential updating. Both household advisers and mutual fund managers respond more strongly closer to the disaster (Tables I.1, 3, and G.5), while the response is more persistent among household advisers, remaining positive through six quarters (Table I.2). These patterns are consistent with documented salience effects of disasters. Alok et al. [2020] find that nearby mutual fund managers underweight disaster-zone stocks more than distant managers, with this aversion declining over time and distance. Gallagher [2014] finds that flood insurance take-up spikes after a flood and then declines to baseline, while take-up in nonflooded communities in the same television market rises at one third the rate. If salience is important, these differences in responses can matter for aggregation. Even under national coverage, investors closer to a disaster may react more strongly to the same news. This can change who the marginal buyer is and how investors who would usually accommodate the flow behave, because those investors may also increase their own demand for green assets.

6.2 Taste or Belief shock

Changes in portfolio allocations can reveal preferences or beliefs shocks. In the previous sections, I discussed whether and whose beliefs or preferences are revealed by holdings in sustainable assets. Here, I focus on households and provide suggestive evidence on the mechanism through which extreme weather events and their interpretation affect portfolios. Household portfolios tend to be inert and therefore reflect beliefs only weakly [Giglio et al., 2021]. Since I found evidence of sizeable changes in portfolios, I now address what might trigger these changes and whether the response resembles a shock to tastes (attitudes) or a shock to climate concern.

Survey evidence supports a belief-based interpretation centered on perceived climate harm, with weaker evidence for beliefs about human causes and climate-policy attitudes. Figure 6A shows that, comparing media markets within the same year, advisers hold larger green ETF shares where more residents expect global warming to harm Americans already or within ten years. Associations with human-causation beliefs and regulatory support are positive but imprecisely estimated.

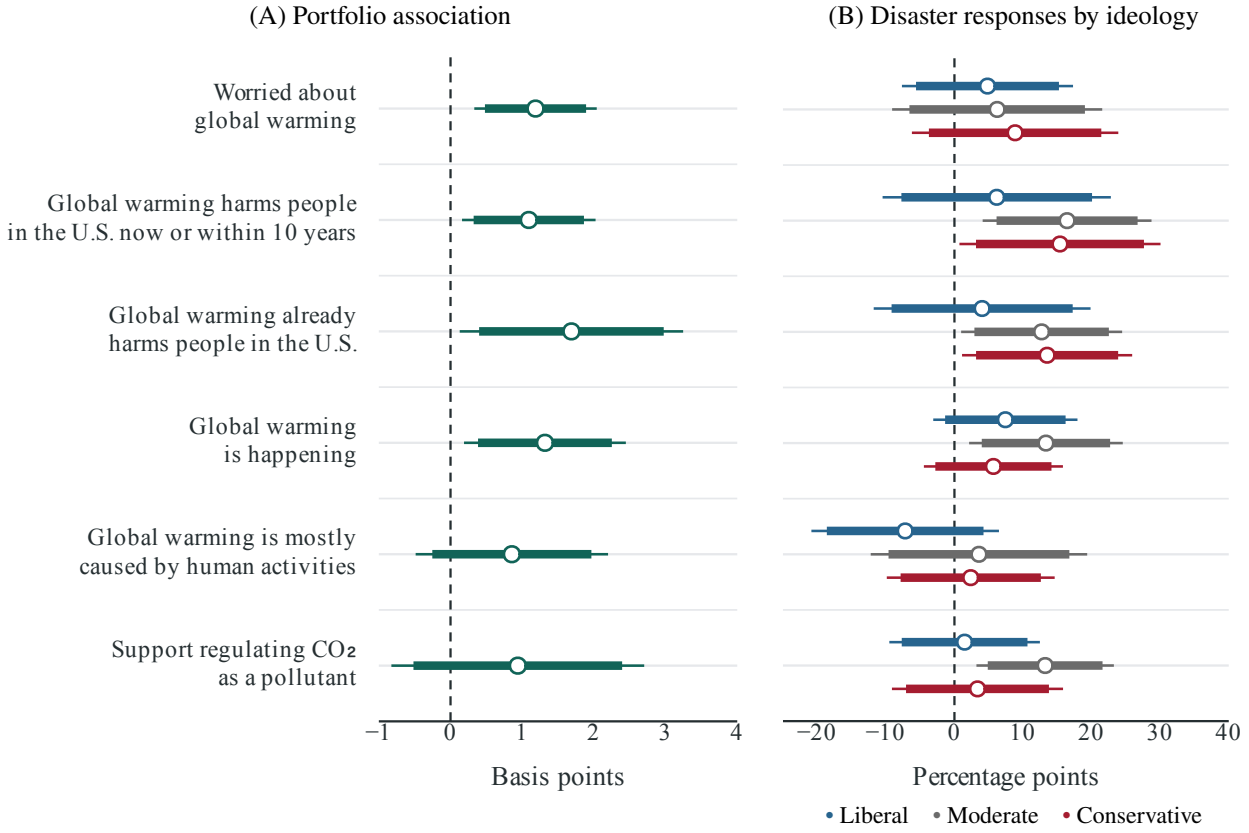
Perceived harm also responds to disaster experience. Following floods, moderates and conservatives become more likely to report harm occurring now or within ten years. Among conservatives, estimated changes in human-causation beliefs and regulatory support are much smaller and statistically insignificant. This contrast indicates that the perceived immediacy of climate consequences connects disaster experience to green investment, while changes in causal beliefs and policy attitudes receive less consistent support.

TABLE 4: PORTFOLIO RESPONSE TO INSTRUMENTED INTERPRETATION

	Portfolio response (basis points)			
	Pre-event prices		Current prices	
	(1)	(2)	(3)	(4)
Panel A. First stage				
Outlet level	−2.29***	−2.29***	−2.29***	−2.29***
	(.50)	(.50)	(.50)	(.50)
County level (transported)	−8.81***	−8.82***	−8.81***	−8.82***
	(2.89)	(2.89)	(2.89)	(2.89)
Newspaper-quarters	6,711	6,711	6,711	6,711
County-events	694	694	694	694
First-stage F, outlet level	20.85	20.85	20.85	20.85
First-stage F, county level	9.32	9.33	9.32	9.33
Panel B. County ETF holdings				
Interpretation (IV)	22.75	22.47	24.96*	24.60*
	(14.30)	(14.21)	(14.69)	(14.59)
Observations	694	694	694	694
Anderson–Rubin p	.064	.067	.051	.054
AR 90% set	[2.5, 61.9]	[2.3, 61.2]	[3.9, 64.7]	[3.6, 63.9]
Panel C. Adviser ETF holdings				
Interpretation (IV)	19.34*	19.18*	22.54**	22.38**
	(10.51)	(10.53)	(11.30)	(11.31)
Observations	4,342	4,342	4,342	4,342
County-events	660	660	660	660
First-stage F	9.43	9.44	9.43	9.44
Anderson–Rubin p	.029	.032	.025	.027
AR 90% set	[4.7, 48.4]	[4.5, 48.2]	[6.3, 52.8]	[6.0, 52.5]
Panel D. Mutual fund managers				
Interpretation (IV)	176.56**	209.87**	107.27	164.27*
	(84.17)	(97.94)	(82.25)	(97.44)
Observations	2,052	2,052	2,052	2,052
County-events	136	136	136	136
First-stage F	11.47	11.13	11.47	11.13
Anderson–Rubin p	.062	.043	.190	.073
AR 90% set	[26.5, 350.6]	[47.1, 429.3]	[−31.0, 287.6]	[14.5, 400.3]
Coverage; own-county deaths and damage	✓	✓	✓	✓
Democratic vote share	✓	✓	✓	✓
Audience reach		✓		✓
Fund size, age and turnover (Panel D)	✓	✓	✓	✓
County, quarter and hazard × distance-band FE	✓	✓	✓	✓

Notes. 2SLS estimates using circulation-weighted log newspaper-chain size as the instrument. Outcomes follow Tables 2 (counties), C.2 (advisers) and 3 (mutual fund managers); advisers are weighted by their pre-event county-episode ETF-book share. Panel D uses managers whose offices are all within 50 miles of the disaster. Interpretation effects are basis points per 10-percentage-point increase. Panel A reports the effect of chain size on the interpretation share, in percentage points, per log point at the outlet level (newspaper and calendar-quarter fixed effects) and per standard deviation at the county level. First-stage F is the Kleibergen–Paap rk Wald statistic. Anderson–Rubin inference is robust to weak instruments. Standard errors and first-stage F statistics are clustered by county (in Panel D, the manager's county) and, at the outlet level, by newspaper. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE 6: CLIMATE BELIEFS, GREEN INVESTMENT, AND DISASTER RESPONSES



Notes. Data are from Climate Change in the American Mind, 2009–2019. Panel A relates annual DMA belief shares to the green share of advisers' ETF holdings, with year fixed effects. Advisers are allocated to a DMA if they predominantly serve households and all their known U.S. offices lie within that DMA. Coefficients are basis points per 10-percentage-point higher belief share. Panel B reports post-disaster changes by ideology, in percentage points, using interviews in affected DMAs from 28 days before through 27 days after flood onset. Regressions include ideology indicators, DMA × survey-year, interview year-month and event fixed effects, and demographic and question-wording controls.

Binary outcomes equal one for the following answers: *Worried:* "How worried are you about global warming?" — somewhat or very worried. *Harm timing:* "When do you think global warming will start to harm people in the United States?" — "In 10 years" or "They are being harmed now"; the second indicator counts only the latter. *Warming happening:* "Do you think that global warming is happening?" and "How sure are you that global warming is happening?" — yes, with somewhat, very or extremely sure. *Human causation:* "Assuming global warming is happening, do you think it is. . . " — caused mostly by human activities. *Regulation:* "How much do you support or oppose the following policies?" — somewhat or strongly support "Regulate carbon dioxide (the primary greenhouse gas) as a pollutant."

Other valid answers equal zero. Thick and thin lines show 90% and 95% confidence intervals, respectively; standard errors are clustered by DMA.

Flood insurance. While the survey evidence documents a correlation between green investment and perceived climate harm, and shows the effect of disasters on the latter, it does not establish that these disaster-induced beliefs actually translate into financial behavior. First-time voluntary flood-insurance purchases provide more concrete evidence that extreme weather events affect households' financial decisions through the climate risk-perception channel, since insurance decisions are not governed by sustainability preferences.

I compare these purchases in affected counties with those in unaffected counties of the same states, separately after floods and wildfires.[52] This comparison helps distinguish interpretation from salience. Floods make flood risk salient directly; wildfires allow me to examine whether climate attribution is associated with demand for protection against another hazard.

By month three, the estimated increase relative to unaffected counties is 41 versus 15 policies per 100,000 residents at the 75th and 25th percentiles of climate attribution after wildfires, compared with 43 versus 40 after floods. The flood response is consistent with salience and experiential learning [Gallagher, 2014], while the cross-hazard insurance response to wildfires suggests that climate interpretation matters for households' demand for protection beyond disaster salience.

6.3 Return Expectations

Affected advisers may buy green ETFs because they expect higher returns [Giglio et al., 2023]. Since a single local disaster event provides no information about either climate risk trends or green firms' cash flows, the return expectation motive requires anticipating that many investors will buy these products as a result of the event[53] and push up their prices. These expectations would be shared by affected and unaffected advisers alike, regardless of their location. The common component of this response is therefore absorbed by the episode fixed effects. For return expectations to explain the stronger response of affected and nearby advisers, and the interpretation gradient among them, affected advisers in higher-interpretation places would need to expect a stronger reaction from others. In particular, they would need to expect a stronger reaction than unaffected advisers do, and than affected advisers in places where the disaster was interpreted less as a climate event. I therefore test whether the response to local interpretation remains after accounting for national attention and national interpretation, and whether either measure changes its strength. I add national coverage and national framing, alone and interacted with local framing, to the county and adviser regressions of Section 3 (Table OA.C.1). The local interpretation response among affected counties and advisers remains positive after accounting for national news. If anything, national attention strengthens this response,[54] while national framing does not significantly moderate this response. National attention widens the response gap between affected places with high and low local interpretation, consistent with polarization of allocation responses.

Finally, if advisers instead bought green ETFs to chase past returns, the response would be larger after green ETFs had done well. The response is the same after strong and weak years for green ETFs (Table I.7).

[52] I estimate local projections of cumulative first-time voluntary NFIP issuance per 100,000 residents from the disaster month through month h. County, episode and state-by-event-year fixed effects absorb persistent county differences and shared shocks and trends, so the identifying variation comes from how insurance purchases vary with climate attribution in affected relative to unaffected counties, conditional on total disaster and climate-only coverage. Standard errors are clustered by county.

[53] Morris and Shin [2002] show how public information can influence behavior through expectations of others' actions. Schmidt-Engelbertz and Vasudevan [2025] document speculation motivated by beliefs about other investors, even when it conflicts with investors' own valuations.

[54] The strengthening is statistically significant for counties and book-weighted advisers, and remains positive but is not statistically significant with equal adviser weights.

After ruling out revaluation, liquidation and the advisers' own incentives (Section 3.4), I can conclude that the observed portfolio responses reflect sustainability demand shocks. Under mild assumptions on the form of return expectations, that expectations about other investors' demand follow national coverage and reading and past green returns and not the local reading of the event, I can also rule out return expectations as the key driver (Section 6.3). What remains to be explained is whether the salience- and interpretation-induced shift in the demand for sustainable assets comes from an ESG taste or beliefs/hedging shock. The suggestive evidence presented in this section points to at least some part played by beliefs: harm perceptions rise after floods among moderates and even conservatives, advisers hold more green ETFs where residents expect harm sooner, and insurance take-up after wildfires rises with the climate interpretation of the event. Future work will tie the shift more tightly to one or the other. Although the origin of the shift is of academic interest, it does not change the role of interpretation, nor its implications in the aggregate.

7 Interpretation, Aggregate Flows and Relative Prices of Green Stocks

At the aggregate level, existing evidence links disasters to increased flows into green ETFs and to the relative outperformance of sustainable assets [Fiordelisi et al., 2023, Misev and Balles, 2024].

To complement this evidence, I compare how flows into ESG-labelled funds relative to other funds respond to national disaster-attention surprises at different levels of national climate interpretation. Attention is measured through national newspaper coverage, and interpretation is the share of disaster articles linking events to climate change. The analysis uses my sample of US domestic-equity mutual funds and ETFs over 2015–2024.[55]

At the average national interpretation share of 25.6%, a one-standard-deviation national disaster-attention surprise is associated with an estimated cumulative increase in flows into ESG-labelled funds relative to other funds equivalent to approximately $288 million over the attention month and the following two months. Figure 7 compares the responses to the same attention surprise at the 25th and 75th percentiles of the national interpretation share, corresponding to 17.9% and 33.9% of coverage. The estimated three-month response is approximately $580 million larger at the 75th than at the 25th percentile.[56]

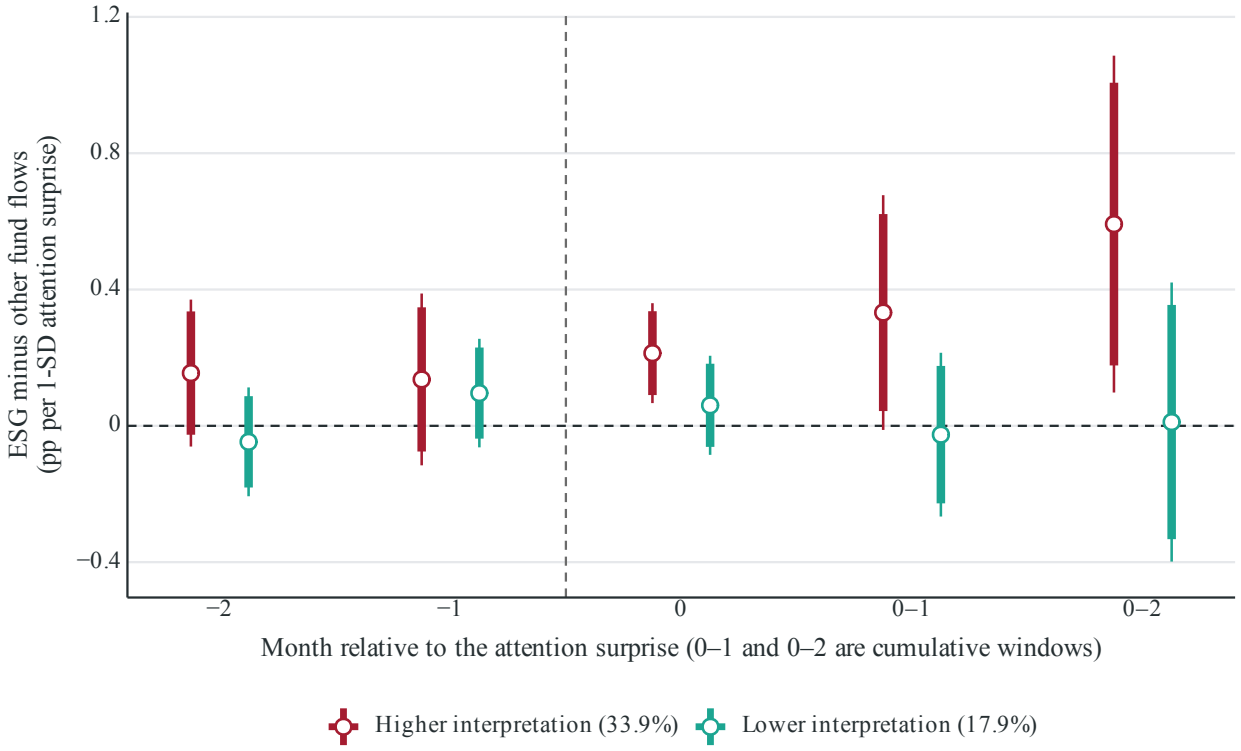
Van der Beck [2026], in turn, shows that ESG flows and, more generally, institutional ESG demand are important drivers of green-minus-nongreen stock returns. Together, these findings motivate asking how much additional aggregate ESG demand and relative appreciation could arise when investors interpret salient disasters more strongly as manifestations of climate change. I examine this question through a counterfactual that applies the locally estimated interpretation response to a broader investor population and uses the calibrated market response to quantify the resulting price adjustment (Appendix J).

I consider an increase in local climate framing from its 25th to its 75th percentile in my sample, holding newspaper coverage at its median and the remaining controls at their sample averages. The estimated additional ESG allocation is 32 basis points over the event quarter and the following quarter. To interpret this response as a demand shift at initial prices, I assume that, conditional on the specification, the comparison

[55] The specification, adapted from Hartzmark and Sussman [2019], compares ESG-labelled funds with other funds within the same investment category and calendar month through investment-category-by-month fixed effects. It controls for fund characteristics and weights observations by log lagged AUM.

[56] In a back-of-the-envelope calculation following Hartzmark and Sussman [2019], I multiply each month's estimated relative flow response by each ESG fund's preceding-month assets and sum across funds and the three response months. The reported amounts average these three-month totals across the 118 eligible starting dates. The response at average interpretation is imprecisely estimated: its 90% and 95% confidence intervals include zero.

Figure 7: National attention and fund flows: log-AUM weights



Notes. This figure plots the adjusted flow-rate responses of ESG-labelled funds relative to other funds to the same positive one-standard-deviation national disaster-attention surprise, evaluated at the 25th and 75th percentiles of national interpretation (17.9% and 33.9%). The sample pools US domestic-equity mutual funds and ETFs over 2015–2024, with log lagged-AUM regression weights. Monthly percentage flows are return-adjusted net flows divided by preceding-month assets. Outcomes at −2, −1 and 0 are monthly rates; 0–1 and 0–2 sum monthly rates over two and three months. The specifications include investment-category-by-month fixed effects and lagged fund size, age, returns and expenses. Thin and thick bars show 95% and 90% confidence intervals, based on monthly-score Newey–West covariance estimates with 12 lags. The −2 and −1 points describe associations with past flows conditional on origin-time controls. Samples vary across outcome windows.

groups have the same average responsiveness of ESG allocations to relative fund prices, so that their responses to the common price movement cancel from the comparison.

The evidence presented in Section 3 that relative flows into ESG-labelled mutual funds also respond to interpretation supports extending the calculation from ETFs to mutual funds and ETFs together. The evidence on rebalancing towards greener stocks by mutual-fund managers without ESG mandates motivates a further extension to institutional investors' equity demand. The central assumption is that investors activated in my counterfactual analyses would respond to stronger climate interpretation as the disaster-exposed investors in my sample do, at the specified level of attention and information environment.

Assume additional ESG purchases are financed by reducing other holdings. Accounting for the ESG holdings already contained in the broad market portfolio, the fund-sector calculation implies an additional ESG-versus-benchmark reallocation of approximately 0.320% of capital. I apply this rate to the $24.74 trillion of institutional equity holdings covered by the calibration, counting the underlying capital once.

Purchasing the ESG portfolio and selling the broad market portfolio generates net purchases of stocks overweighted by ESG funds and net sales of underweighted stocks. I translate this demand into valuations using van der Beck's [2026] ESG flow multiplier, $\kappa_{\text{ESG}} \simeq 0.72$, which measures the difference in green and nongreen stocks' dollar valuation changes per dollar reallocated. Expressing the same response in return units gives $M_{\text{ESG}} \simeq 1.12$, relating the green-minus-nongreen return to relative demand normalized by each group's market capitalization.[57] This way I can incorporate into my counterfactual the investors' willingness to accommodate trades as prices change, under the assumption of comparable portfolio demand and market accommodation.

To compare with annual returns, consider disasters occurring throughout a year that collectively activate the institutional investor base,[58] with each investor completing the estimated additional adjustment once during the year. I compare their reallocations under stronger and weaker climate interpretation, holding attention and national news fixed. Under these assumptions, stronger interpretation generates $79.21 billion of additional ESG-versus-benchmark demand, implying $56.81 billion in additional green-minus-nongreen valuation change and 34 basis points of additional return differential. This additional appreciation is equivalent in magnitude to 18.2% of the average annual return contribution attributed to institutional ESG demand and 15.2% of the observed annual return spread.

8 Conclusion

This paper measures, for each disaster episode and county, the news attention the event receives as well as, separately, the focus and narratives provided, especially on its climate change interpretation. I find that the interpretation sets the direction of investors' portfolio responses. Among affected counties, higher climate interpretation raises the green-ETF share of household portfolios managed by local advisers, net inflows into ESG mutual funds relative to non-ESG funds rise among funds headquartered in affected states, and managers of mutual funds without an ESG mandate raise their active tilt toward greener firms with the interpretation reaching their office. I confirm the effect of locally supplied interpretation narratives using changes in the ownership of the county's outlets, which lower the climate-interpreted share of local disaster coverage while holding coverage constant, and survey and insurance evidence points to perceived climate

[57] Both multipliers are calibrated using van der Beck's [2026] December 2022 inputs.

[58] See Appendix J for the same exercise applied to demand for (i) green ETFs alone and (ii) ESG-labelled mutual funds and ETFs together.

harm as the link between the experience and the investment. In a counterfactual in which disasters activate the institutional investor base, an interquartile increase in climate interpretation implies 34 basis points of ESG-stock appreciation relative to other stocks, 15.2% of the observed annual return spread.

The analysis is agnostic about whether the interpretation moves portfolios by inducing hedging of climate risk or through tastes for sustainable assets. Beyond the scope of this paper is a demand system that incorporates the interpretations and portfolio holdings of the different types of local and institutional investors, with their elasticities, to assess whose reading of an event is reflected in asset prices. Climate disasters provide the cleanest setting for the measurement, since the event is unannounced and its coverage is local, but the approach extends to other local shocks that local news interprets, such as a spike in local prices interpreted as transitory or persistent, or a plant closure interpreted as cyclical or structural. Professional forecasters disagree about the path of the federal funds rate at every horizon (Andrade, Crump, Eusepi, and Moench, 2016). More work is needed to understand whether the local interpretation of such shocks accounts for part of this disagreement, and whether investors who receive one interpretation respond differently to the monetary-policy announcements that follow.

Because of the increasing incidence of natural disasters due to climate change, the findings should be of interest to regulators, academics, and investors asking how climate risk reaches asset prices. Prior work prices the firms exposed to the event, with nearby managers underweighting disaster-zone stocks (Alok, Kumar, and Wermers, 2020) and options markets pricing the uncertainty of firms in a hurricane's path (Kruttli, Roth Tran, and Watugala, 2025). The results here add a route that runs through the reading of the event. A disaster read as climate change moves demand toward sustainable assets, a category that spans firms and places far from the damage and not even indirectly affected, so part of the price footprint of a disaster forms away from the exposed firms and depends on the interpretation the event receives.

Finally, the level of journalistic resources dedicated to local news is in long-term decline in the United States, through the closure of newspapers and the consolidation of those that survive (Gao, Lee, and Murphy, 2020; Ewens, Gupta, and Howell, 2022; Martin and McCrain, 2019), and the cuts now reach climate coverage directly.[59] Outlets acquired by a chain cover the same disaster with a lower climate-interpreted share, and the households and managers who receive that coverage respond less. The results shed light on a further consequence of media consolidation: the loss of climate interpretation and discourse around extreme weather events across the places that experience them; how far this trend would weaken the response of investors toward sustainability is a question that the measure introduced here can be used to study further.

[59]Newspaper closures raise municipal borrowing costs (Gao, Lee, and Murphy, 2020); under private-equity ownership, coverage shifts away from local governance (Ewens, Gupta, and Howell, 2022); and after conglomerate acquisitions, local television news moves toward national politics (Martin and McCrain, 2019). In 2026 the Washington Post cut the majority of its climate team in a restructuring, NPR closed its stand-alone climate desk after the withdrawal of federal funding, and CBS News dismantled its climate team after the change of ownership of its parent: Yale Climate Connections, February 2026, https://yaleclimateconnections.org/2026/02/elegy-for-the-washington-post-climate-team/; NPR Public Editor, June 2026, https://www.npr.org/sections/npr-public-editor/2026/06/11/g-s1-127514/how-will-npr-layoffs-affect-climate-stories; Nieman Reports, https://niemanreports.org/climate-journalism-cuts-layoffs-environmental-natural-disasters-reporters-global-warming/; Media Matters counts 87 minutes of CBS climate coverage over January–September 2026 against 222 over the same months of 2025, https://www.mediamatters.org/broadcast-networks/under-bari-weiss-cbs-news-dismantled-its-climate-team-and-retreated-climate.

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